

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6323HUX (18 x 10 feet) - 7.82 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,149.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.9 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.115	Water Loading	6.72 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	34.14
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.997	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	43,356.5 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.947	Dry Air Rate	2,856.9 lb/min/cell
L/G Ratio	1.816	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	8.4 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	397.4	0.0692	14.80	0.0091
Rain Zone	171.0	494.6	0.0692	14.80	0.0249
Fill	171.0	500.8	0.0687	14.99	0.1779
Spray Zone	171.0	507.1	0.0683	14.99	0.0708
Drift Eliminator	171.0	507.1	0.0683	15.18	0.0271
Fan Inlet	57.9	1,497.6	0.0683	15.18	0.0636
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3724
Net Fan VP	57.9	1,497.6	0.0683	15.18	0.1272
Fan Total Pressure					0.4995
Stack Exit	59.7	1,451.7	0.0683	15.18	0.1195

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7G2DOL1 (12 x 12 feet, 5 fill layer) - 9.06 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 12.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,174.8 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 12.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 12.00 ft
Cell Width = 12.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 9.00 ft
Seal Disk/Hub Diameter = 1.62 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 9.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.3 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.125	Water Loading	8.59 gpm/ft ²
CF-1900SB - 5.00 ft	1.112	Pressure Ratio	46.26
Fill Total	1.112	L/G * KaV/L	2.7
Rain Zone	0.059	Fan:Box Ratio	45.8 %
Total KaV/L (CTI)	1.296	Fan Coverage	96.1 %
KaV/L Derate (%)	5	Airflow @ Fan	74,476.3 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	61.72 %
KaV/L Adjusted	1.231	Dry Air Rate	4,881.2 lb/min/cell
L/G Ratio	2.220	Total Fill Ht.	5 ft
		Fill Area	136.8 ft ² /cell
		Evaporation Rate	0.68 %
		Evaporation Rate	8.8 gpm (tower)
		Inlet DBT	91.7 °F
		Exit WBT	95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	452.9	0.0691	14.81	0.0118
Rain Zone	136.8	528.4	0.0691	14.81	0.0285
Fill	136.8	536.4	0.0686	15.03	0.3572
Spray Zone	136.8	544.4	0.0681	15.03	0.1156
Drift Eliminator	136.8	544.4	0.0681	15.26	0.0316
Fan Inlet	63.9	1,164.9	0.0681	15.26	0.0383
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.5817
Net Fan VP	63.9	1,164.9	0.0681	15.26	0.0767
Fan Total Pressure					0.6584
Stack Exit	66.0	1,128.5	0.0681	15.26	0.0720

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7F2DOL1 (11 x 11 feet, 5 fill layer) - 7 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 890.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft

Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 11.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 11.00 ft
Cell Width = 11.00 ft
Net Total Inlet Area = 146.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 7.00 ft
Seal Disk/Hub Diameter = 1.26 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 7.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 112.7 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.116	Water Loading 7.74 gpm/ft ²
CF-1900SB - 5.00 ft 1.115	Pressure Ratio 54.13
Fill Total 1.115	L/G * KaV/L 2.7
Rain Zone 0.058	Fan:Box Ratio 33.3 %
Total KaV/L (CTI) 1.289	Fan Coverage 92.5 %
KaV/L Derate (%) 5	Airflow @ Fan 57,296.2 cfm/cell
HWT Correction (%) 0.0	Effective Fan Eff. 56.79 %
KaV/L Adjusted 1.225	Dry Air Rate 3,755.6 lb/min/cell
L/G Ratio 2.213	Total Fill Ht. 5 ft
	Fill Area 115.0 ft ² /cell
	Evaporation Rate 0.68 %
	Evaporation Rate 6.8 gpm (tower)
	Inlet DBT 91.7 °F
	Exit WBT 95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	146.3	380.2	0.0691	14.81	0.0083
Rain Zone	115.0	483.8	0.0691	14.81	0.0245
Fill	115.0	491.1	0.0686	15.03	0.2948
Spray Zone	115.0	498.4	0.0681	15.03	0.0954
Drift Eliminator	115.0	498.4	0.0681	15.26	0.0260
Fan Inlet	39.1	1,465.7	0.0681	15.26	0.0607
Regain					0.0000
Bouyancy					-0.0012
Sum Static dp					0.5085
Net Fan VP	39.1	1,465.7	0.0681	15.26	0.1214
Fan Total Pressure					0.6299
Stack Exit	40.3	1,420.4	0.0681	15.26	0.1140

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7E2DOL1 (10 x 10 feet, 5 fill layer) - 6.86 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 804.6 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 10.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 10.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 133.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 7.00 ft
Seal Disk/Hub Diameter = 1.26 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 7.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 1.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.3 %

KaV/L Data

Spray Zone	0.125
CF-1900SB - 5.00 ft	1.112
Fill Total	1.112
Rain Zone	0.059
Total KaV/L (CTI)	1.296
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.232
L/G Ratio	2.220

Thermal Data - Misc.

Water Loading	8.47 gpm/ft ²
Pressure Ratio	66.23
L/G * KaV/L	2.7
Fan:Box Ratio	40.3 %
Fan Coverage	86.1 %
Airflow @ Fan	51,943.5 cfm/cell
Effective Fan Eff.	56.79 %
Dry Air Rate	3,404.4 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	95.0 ft ² /cell
Evaporation Rate	0.68 %
Evaporation Rate	6.2 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	133.0	379.1	0.0691	14.81	0.0082
Rain Zone	95.0	530.7	0.0691	14.81	0.0287
Fill	95.0	538.7	0.0686	15.03	0.3606
Spray Zone	95.0	546.8	0.0681	15.03	0.1167
Drift Eliminator	95.0	546.8	0.0681	15.26	0.0319
Fan Inlet	39.1	1,328.7	0.0681	15.26	0.0499
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.5951
Net Fan VP	39.1	1,328.7	0.0681	15.26	0.0998
Fan Total Pressure					0.6948
Stack Exit	40.3	1,287.7	0.0681	15.26	0.0937

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7C3DOL1 (8 x 8 feet, 5 fill layer) - 5.93 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 7.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 498.4 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 8.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 8.00 ft
Cell Width = 8.00 ft
Net Total Inlet Area = 106.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 1.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.2 %

KaV/L Data

Spray Zone	0.118
CF-1900SB - 5.00 ft	1.114
Fill Total	1.114
Rain Zone	0.058
Total KaV/L (CTI)	1.290
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.226
L/G Ratio	2.215

Thermal Data - Misc.

Water Loading	8.2 gpm/ft ²
Pressure Ratio	101.68
L/G * KaV/L	2.7
Fan:Box Ratio	46.7 %
Fan Coverage	95.5 %
Airflow @ Fan	30,790.3 cfm/cell
Effective Fan Eff.	35.87 %
Dry Air Rate	2,018.2 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	60.8 ft ² /cell
Evaporation Rate	0.68 %
Evaporation Rate	3.6 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	106.4	280.9	0.0691	14.81	0.0045
Rain Zone	60.8	491.6	0.0691	14.81	0.0252
Fill	60.8	499.0	0.0686	15.03	0.3051
Spray Zone	60.8	506.4	0.0681	15.03	0.0988
Drift Eliminator	60.8	506.4	0.0681	15.26	0.0269
Fan Inlet	29.0	1,063.5	0.0681	15.26	0.0320
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.4914
Net Fan VP	29.0	1,063.5	0.0681	15.26	0.0639
Fan Total Pressure					0.5553
Stack Exit	29.9	1,030.9	0.0681	15.26	0.0601

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7I4DO1 (14 x 14 feet, 4 fill layer) - 14.87 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 20.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,780.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 3.92 ft
Total Fill Height = 3.92 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 8.67 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 199.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.83 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 110.3 %

KaV/L Data

Spray Zone	0.119
CF-1200MABT - 3.92 ft	1.150
Fill Total	1.150
Rain Zone	0.058
Total KaV/L (CTI)	1.328
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.261
L/G Ratio	2.249

Thermal Data - Misc.

Water Loading	8.33 gpm/ft ²
Pressure Ratio	33.59
L/G * KaV/L	2.8
Fan:Box Ratio	36.1 %
Fan Coverage	85.0 %
Airflow @ Fan	110,532.4 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	7,240.6 lb/min/cell
Total Fill Ht.	3.92 ft
Fill Area	213.8 ft ² /cell
Evaporation Rate	0.68 %
Evaporation Rate	13.3 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	95.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	199.5	537.5	0.0691	14.81	0.0166
Rain Zone	213.8	501.7	0.0691	14.81	0.0264
Fill	213.8	509.4	0.0686	15.04	0.3817
Spray Zone	213.8	517.1	0.0680	15.04	0.1042
Drift Eliminator	213.8	517.1	0.0680	15.27	0.0282
Fan Inlet	78.6	1,405.6	0.0680	15.27	0.0558
Regain					0.0000
Bouyancy					-0.0014
Sum Static dp					0.6114
Net Fan VP	78.6	1,405.6	0.0680	15.27	0.1116
Fan Total Pressure					0.7231
Stack Exit	81.2	1,361.6	0.0680	15.27	0.1047

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R3)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height 105% Performance targetComments:Thermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 700.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 0.75 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 2.75 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 5.46 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft
No. of Cells = 3
Inlet Height = 2.75 ft
Cell Length = 5.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 109.7 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft
Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.11 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.79 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 9.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 97.6 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	8.19 gpm/ft ²
CF-1200MABT - 1.96 ft	0.821	Pressure Ratio	33.41
Fill Total	0.821	L/G * KaV/L	1.7
Rain Zone	0.053	Fan:Box Ratio	44.2 %
Total KaV/L (CTI)	0.988	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	15,926.5 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	41.88 %
KaV/L Adjusted	0.939	Dry Air Rate	1,049.8 lb/min/cell
L/G Ratio	1.798	Total Fill Ht.	1.96 ft
		Fill Area	28.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.6 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	109.7	424.9	0.0692	14.80	0.0104
Rain Zone	85.5	545.2	0.0692	14.80	0.0187
Fill	85.5	552.0	0.0687	14.99	0.2188
Spray Zone	85.5	558.8	0.0683	14.99	0.0651
Drift Eliminator	85.5	558.8	0.0683	15.17	0.0336
Fan Inlet	38.5	1,239.5	0.0683	15.17	0.0436
Regain					0.0000
Bouyancy					-0.0008
Sum Static dp					0.3892
Net Fan VP	38.5	1,239.5	0.0683	15.17	0.0871
Fan Total Pressure					0.4763
Stack Exit	39.8	1,201.4	0.0683	15.17	0.0819

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 42-32-28c 3000LPMThermal Conditions

Tower Model = DFC 70UX
Power = 12.50 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 783.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 18.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 107.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 12.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 12.00 ft
Cell Width = 12.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 9.00 ft
Seal Disk/Hub Diameter = 1.62 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 9.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 104.3 %

KaV/L Data

Spray Zone	0.143
CF-1200MABT - 2.94 ft	1.444
Fill Total	1.444
Rain Zone	0.068
Total KaV/L (CTI)	1.656
KaV/L Derate (%)	5
HWT Correction (%)	2.3
KaV/L Adjusted	1.535
L/G Ratio	1.319

Thermal Data - Misc.

Water Loading	5.72 gpm/ft ²
Pressure Ratio	38.00
L/G * KaV/L	2.0
Fan:Box Ratio	45.8 %
Fan Coverage	96.5 %
Airflow @ Fan	79,311.5 cfm/cell
Effective Fan Eff.	61.72 %
Dry Air Rate	5,121.9 lb/min/cell
Total Fill Ht.	2.94 ft
Fill Area	136.8 ft ² /cell
Evaporation Rate	1.57 %
Evaporation Rate	12.8 gpm (tower)
Inlet DBT	91.0 °F
Exit WBT	99.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	474.3	0.0692	14.78	0.0129
Rain Zone	136.8	553.4	0.0692	14.78	0.0194
Fill	136.8	566.6	0.0683	15.13	0.3280
Spray Zone	136.8	579.8	0.0674	15.13	0.0953
Drift Eliminator	136.8	579.8	0.0674	15.48	0.0359
Fan Inlet	63.9	1,240.5	0.0674	15.48	0.0431
Regain					0.0000
Bouyancy					-0.0024
Sum Static dp					0.5321
Net Fan VP	63.9	1,240.5	0.0674	15.48	0.0861
Fan Total Pressure					0.6183
Stack Exit	66.0	1,201.8	0.0674	15.48	0.0808

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7I5DOL1 (15 x 15 feet, 7 fill layer) - 14.28 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 20.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,815.6 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 7 ft
Rain Height = 3.50 ft

Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 199.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 16.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.7 %

KaV/L Data

Spray Zone	0.114
CF-1900SB - 7.00 ft	1.371
Fill Total	1.371
Rain Zone	0.057
Total KaV/L (CTI)	1.542
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.465
L/G Ratio	2.402

Thermal Data - Misc.

Water Loading	8.49 gpm/ft ²
Pressure Ratio	38.13
L/G * KaV/L	3.5
Fan:Box Ratio	36.1 %
Fan Coverage	83.9 %
Airflow @ Fan	107,121.0 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	6,998.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	213.8 ft ² /cell
Evaporation Rate	0.67 %
Evaporation Rate	13.6 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	96.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	199.5	519.5	0.0691	14.81	0.0155
Rain Zone	213.8	484.9	0.0691	14.81	0.0264
Fill	213.8	493.0	0.0685	15.06	0.4215
Spray Zone	213.8	501.2	0.0679	15.06	0.1009
Drift Eliminator	213.8	501.2	0.0679	15.31	0.0262
Fan Inlet	78.6	1,362.3	0.0679	15.31	0.0523
Regain					0.0000
Bouyancy					-0.0015
Sum Static dp					0.6414
Net Fan VP	78.6	1,362.3	0.0679	15.31	0.1047
Fan Total Pressure					0.7461
Stack Exit	81.2	1,319.6	0.0679	15.31	0.0982

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7I4DOL1 (15 x 15 feet, 6 fill layer) - 13.61 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 20.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,780.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 6 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 199.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.6 %

KaV/L Data

Spray Zone	0.118
CF-1900SB - 6.00 ft	1.247
Fill Total	1.247
Rain Zone	0.058
Total KaV/L (CTI)	1.422
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.351
L/G Ratio	2.325

Thermal Data - Misc.

Water Loading	8.33 gpm/ft ²
Pressure Ratio	34.05
L/G * KaV/L	3.1
Fan:Box Ratio	36.1 %
Fan Coverage	83.9 %
Airflow @ Fan	110,240.5 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	7,211.8 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	213.8 ft ² /cell
Evaporation Rate	0.67 %
Evaporation Rate	13.6 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	96.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	199.5	535.4	0.0691	14.81	0.0164
Rain Zone	213.8	499.7	0.0691	14.81	0.0270
Fill	213.8	507.7	0.0685	15.05	0.3831
Spray Zone	213.8	515.7	0.0680	15.05	0.1055
Drift Eliminator	213.8	515.7	0.0680	15.29	0.0280
Fan Inlet	78.6	1,401.9	0.0680	15.29	0.0555
Regain					0.0000
Bouyancy					-0.0014
Sum Static dp					0.6140
Net Fan VP	78.6	1,401.9	0.0680	15.29	0.1109
Fan Total Pressure					0.7250
Stack Exit	81.2	1,358.0	0.0680	15.29	0.1041

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7I3DOL1 (15 x 15 feet, 5 fill layer) - 13.33 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 20.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,744.4 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 199.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 114.4 %

KaV/L Data

Spray Zone	0.123
CF-1900SB - 5.00 ft	1.113
Fill Total	1.113
Rain Zone	0.059
Total KaV/L (CTI)	1.294
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.230
L/G Ratio	2.218

Thermal Data - Misc.

Water Loading	8.16 gpm/ft ²
Pressure Ratio	29.86
L/G * KaV/L	2.7
Fan:Box Ratio	36.1 %
Fan Coverage	83.9 %
Airflow @ Fan	113,838.2 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	7,461.2 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	213.8 ft ² /cell
Evaporation Rate	0.68 %
Evaporation Rate	13.5 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	199.5	553.9	0.0691	14.81	0.0176
Rain Zone	213.8	516.9	0.0691	14.81	0.0274
Fill	213.8	524.8	0.0686	15.03	0.3404
Spray Zone	213.8	532.6	0.0681	15.03	0.1102
Drift Eliminator	213.8	532.6	0.0681	15.26	0.0301
Fan Inlet	78.6	1,447.7	0.0681	15.26	0.0592
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.5836
Net Fan VP	78.6	1,447.7	0.0681	15.26	0.1185
Fan Total Pressure					0.7021
Stack Exit	81.2	1,402.3	0.0681	15.26	0.1111

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7H5GOL1 (14 x 14 feet, 7 fill layer) - 13.07 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 20.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,637.6 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 7 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 186.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 16.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 114.8 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 7.00 ft	1.369
Fill Total	1.369
Rain Zone	0.057
Total KaV/L (CTI)	1.547
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.470
L/G Ratio	2.405

Thermal Data - Misc.

Water Loading	8.79 gpm/ft ²
Pressure Ratio	44.07
L/G * KaV/L	3.5
Fan:Box Ratio	41.4 %
Fan Coverage	90.3 %
Airflow @ Fan	99,227.3 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	6,482.2 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	186.2 ft ² /cell
Evaporation Rate	0.67 %
Evaporation Rate	12.6 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	96.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	186.2	515.6	0.0691	14.81	0.0153
Rain Zone	186.2	515.6	0.0691	14.81	0.0294
Fill	186.2	524.2	0.0685	15.06	0.4822
Spray Zone	186.2	532.9	0.0679	15.06	0.1153
Drift Eliminator	186.2	532.9	0.0679	15.31	0.0301
Fan Inlet	78.6	1,261.9	0.0679	15.31	0.0449
Regain					0.0000
Bouyancy					-0.0015
Sum Static dp					0.7156
Net Fan VP	78.6	1,261.9	0.0679	15.31	0.0898
Fan Total Pressure					0.8054
Stack Exit	81.2	1,222.3	0.0679	15.31	0.0843

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7H4DOL1 (14 x 14 feet, 6 fill layer) - 10.71 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 15.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,530.8 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 6 ft
Rain Height = 3.50 ft

Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 186.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.7 %

KaV/L Data

Spray Zone	0.115
CF-1900SB - 6.00 ft	1.247
Fill Total	1.247
Rain Zone	0.058
Total KaV/L (CTI)	1.420
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.349
L/G Ratio	2.323

Thermal Data - Misc.

Water Loading	8.22 gpm/ft ²
Pressure Ratio	38.77
L/G * KaV/L	3.1
Fan:Box Ratio	41.4 %
Fan Coverage	90.3 %
Airflow @ Fan	93,315.3 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	6,104.8 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	186.2 ft ² /cell
Evaporation Rate	0.67 %
Evaporation Rate	11.5 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	96.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	186.2	485.5	0.0691	14.81	0.0135
Rain Zone	186.2	485.5	0.0691	14.81	0.0257
Fill	186.2	493.3	0.0685	15.05	0.3599
Spray Zone	186.2	501.2	0.0680	15.05	0.0991
Drift Eliminator	186.2	501.2	0.0680	15.29	0.0263
Fan Inlet	78.6	1,186.7	0.0680	15.29	0.0397
Regain					0.0000
Bouyancy					-0.0014
Sum Static dp					0.5629
Net Fan VP	78.6	1,186.7	0.0680	15.29	0.0795
Fan Total Pressure					0.6424
Stack Exit	81.2	1,149.5	0.0680	15.29	0.0746

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7H3DOL1 (14 x 14 feet, 5 fill layer) - 10.35 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 15.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,495.2 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 186.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.1 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.120	Water Loading 8.03 gpm/ft ²
CF-1900SB - 5.00 ft 1.114	Pressure Ratio 33.99
Fill Total 1.114	L/G * KaV/L 2.7
Rain Zone 0.059	Fan:Box Ratio 41.4 %
Total KaV/L (CTI) 1.292	Fan Coverage 90.3 %
KaV/L Derate (%) 5	Airflow @ Fan 96,558.1 cfm/cell
HWT Correction (%) 0.0	Effective Fan Eff. 62.87 %
KaV/L Adjusted 1.228	Dry Air Rate 6,328.8 lb/min/cell
L/G Ratio 2.216	Total Fill Ht. 5 ft
	Fill Area 186.2 ft ² /cell
	Evaporation Rate 0.68 %
	Evaporation Rate 11.5 gpm (tower)
	Inlet DBT 91.7 °F
	Exit WBT 95.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	186.2	503.4	0.0691	14.81	0.0145
Rain Zone	186.2	503.4	0.0691	14.81	0.0262
Fill	186.2	511.0	0.0686	15.03	0.3212
Spray Zone	186.2	518.6	0.0681	15.03	0.1040
Drift Eliminator	186.2	518.6	0.0681	15.26	0.0284
Fan Inlet	78.6	1,227.9	0.0681	15.26	0.0426
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.5356
Net Fan VP	78.6	1,227.9	0.0681	15.26	0.0852
Fan Total Pressure					0.6208
Stack Exit	81.2	1,189.4	0.0681	15.26	0.0800

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7H2DOL1 (14 x 14 feet, 5 fill layer) - 8.68 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 12.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,352.8 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 186.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 9.00 ft
Seal Disk/Hub Diameter = 1.62 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 9.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.1 %

KaV/L Data

Spray Zone	0.110
CF-1900SB - 5.00 ft	1.116
Fill Total	1.116
Rain Zone	0.058
Total KaV/L (CTI)	1.285
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.221
L/G Ratio	2.209

Thermal Data - Misc.

Water Loading	7.27 gpm/ft ²
Pressure Ratio	33.57
L/G * KaV/L	2.7
Fan:Box Ratio	33.7 %
Fan Coverage	84.2 %
Airflow @ Fan	87,601.8 cfm/cell
Effective Fan Eff.	61.72 %
Dry Air Rate	5,742.5 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	186.2 ft ² /cell
Evaporation Rate	0.68 %
Evaporation Rate	10.4 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	95.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	186.2	456.7	0.0691	14.81	0.0120
Rain Zone	186.2	456.7	0.0691	14.81	0.0222
Fill	186.2	463.6	0.0686	15.03	0.2606
Spray Zone	186.2	470.5	0.0681	15.03	0.0843
Drift Eliminator	186.2	470.5	0.0681	15.25	0.0228
Fan Inlet	63.9	1,370.2	0.0681	15.25	0.0531
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.4536
Net Fan VP	63.9	1,370.2	0.0681	15.25	0.1061
Fan Total Pressure					0.5597
Stack Exit	66.0	1,327.4	0.0681	15.25	0.0996

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.

Project: DFC 60HUX Models (R3)
Delhi, Delhi
IndiaProject Description: Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height 105% Performance targetComments:Thermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 700.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 0.75 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 2.75 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 5.46 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft
No. of Cells = 3
Inlet Height = 2.75 ft
Cell Length = 5.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 109.7 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft
Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.11 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.79 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 9.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 97.6 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	8.19 gpm/ft ²
CF-1200MABT - 1.96 ft	0.821	Pressure Ratio	33.41
Fill Total	0.821	L/G * KaV/L	1.7
Rain Zone	0.053	Fan:Box Ratio	44.2 %
Total KaV/L (CTI)	0.988	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	15,926.5 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	41.88 %
KaV/L Adjusted	0.939	Dry Air Rate	1,049.8 lb/min/cell
L/G Ratio	1.798	Total Fill Ht.	1.96 ft
		Fill Area	28.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.6 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	109.7	424.9	0.0692	14.80	0.0104
Rain Zone	85.5	545.2	0.0692	14.80	0.0187
Fill	85.5	552.0	0.0687	14.99	0.2188
Spray Zone	85.5	558.8	0.0683	14.99	0.0651
Drift Eliminator	85.5	558.8	0.0683	15.17	0.0336
Fan Inlet	38.5	1,239.5	0.0683	15.17	0.0436
Regain					0.0000
Bouyancy					-0.0008
Sum Static dp					0.3892
Net Fan VP	38.5	1,239.5	0.0683	15.17	0.0871
Fan Total Pressure					0.4763
Stack Exit	39.8	1,201.4	0.0683	15.17	0.0819

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.

Project: DFC 60HUX Models (R3)
Delhi, Delhi
IndiaProject Description: Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height 105% Performance targetComments:Thermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 650.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 15.00 ft
No. of Cells = 3
Inlet Height = 4.00 ft
Cell Length = 5.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft
Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.11 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 104.7 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.122	Water Loading	7.6 gpm/ft ²
CF-1200MABT - 1.96 ft	0.812	Pressure Ratio	76.37
Fill Total	0.812	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	44.2 %
Total KaV/L (CTI)	1.002	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	15,633.1 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	41.88 %
KaV/L Adjusted	0.952	Dry Air Rate	1,029.9 lb/min/cell
L/G Ratio	1.826	Total Fill Ht.	1.96 ft
		Fill Area	28.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.6 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	286.5	0.0692	14.80	0.0047
Rain Zone	85.5	534.9	0.0692	14.80	0.0286
Fill	85.5	541.7	0.0687	14.99	0.2105
Spray Zone	85.5	548.5	0.0683	14.99	0.0841
Drift Eliminator	85.5	548.5	0.0683	15.18	0.0322
Fan Inlet	38.5	1,216.7	0.0683	15.18	0.0420
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.4013
Net Fan VP	38.5	1,216.7	0.0683	15.18	0.0839
Fan Total Pressure					0.4852
Stack Exit	39.8	1,179.3	0.0683	15.18	0.0788

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 180 Twin CellThermal Conditions

Tower Model = DPC 70
Power = 5.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 525.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 10.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 100.00 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 3.92 ft
Total Fill Height = 3.92 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 8.67 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 10.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 5.00 ft
Cell Width = 8.00 ft
Net Total Inlet Area = 119.7 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft

Fan Tip Clearance = 0.67 in
Plenum Hole Diameter = 4.12 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 1.83 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 104.3 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	6.91 gpm/ft ²
CF-1200MABT - 3.92 ft	1.310	Pressure Ratio	89.32
Fill Total	1.310	L/G * KaV/L	2.7
Rain Zone	0.060	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	1.484	Fan Coverage	90.1 %
KaV/L Derate (%)	5	Airflow @ Fan	18,135.4 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	34.50 %
KaV/L Adjusted	1.410	Dry Air Rate	1,181.5 lb/min/cell
L/G Ratio	1.921	Total Fill Ht.	3.92 ft
		Fill Area	38.0 ft ² /cell
		Evaporation Rate	0.89 %
		Evaporation Rate	4.9 gpm (tower)
		Inlet DBT	91.7 °F
		Exit WBT	97.1 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	119.7	292.4	0.0691	14.81	0.0049
Rain Zone	76.0	460.5	0.0691	14.81	0.0199
Fill	76.0	468.9	0.0684	15.08	0.3106
Spray Zone	76.0	477.2	0.0678	15.08	0.0793
Drift Eliminator	76.0	477.2	0.0678	15.35	0.0235
Fan Inlet	25.7	1,409.1	0.0678	15.35	0.0559
Regain					0.0000
Bouyancy					-0.0012
Sum Static dp					0.4928
Net Fan VP	25.7	1,409.1	0.0678	15.35	0.1118
Fan Total Pressure					0.6046
Stack Exit	26.6	1,365.8	0.0678	15.35	0.1050

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 440 TR Low Height (110 X 4 inline) - CasagrandThermal Conditions

Tower Model = DPC 70
Power = 10.00 HP
Cold Water Temperature = 83.00 °F
Flow Rate = 1,250.0 gpm
Wet Bulb Temperature = 78.00 °F
Relative Humidity = 70.00 %

Range = 9.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 92.00 °F
Approach = 5.00 °F

Fill Section

Spray Height = 0.66 ft
CF-1900SB = 2.00 ft
Total Fill Height = 2 ft
Rain Height = 1.64 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 4.30 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 40.00 ft

No. of Cells = 4
Inlet Height = 1.64 ft
Cell Length = 10.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 155.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 7.00 ft
Seal Disk/Hub Diameter = 1.26 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 7.17 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 1.20 ft
Spray to Drift Height = 0.66 ft
Fan Deck Height = 6.16 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 127.2 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.144	Water Loading 3.29 gpm/ft ²
CF-1900SB - 2.00 ft 1.084	Pressure Ratio 3.04
Fill Total 1.084	L/G * KaV/L 1.0
Rain Zone 0.046	Fan:Box Ratio 40.3 %
Total KaV/L (CTI) 1.273	Fan Coverage 71.9 %
KaV/L Derate (%) 5	Airflow @ Fan 59,520.0 cfm/cell
HWT Correction (%) 0.0	Effective Fan Eff. 56.79 %
KaV/L Adjusted 1.210	Dry Air Rate 4,048.5 lb/min/cell
L/G Ratio 0.815	Total Fill Ht. 2 ft
	Fill Area 95.0 ft ² /cell
	Evaporation Rate 0.90 %
	Evaporation Rate 14.4 gpm (tower)
	Inlet DBT 86.2 °F
	Exit WBT 84.7 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	155.8	1,515.0	0.0699	14.58	0.1333
Rain Zone	380.0	621.1	0.0699	14.58	0.0063
Fill	380.0	623.8	0.0699	14.64	0.1735
Spray Zone	380.0	626.5	0.0698	14.64	0.0477
Drift Eliminator	380.0	626.5	0.0698	14.70	0.0439
Fan Inlet	156.4	1,522.6	0.0698	14.70	0.0672
Regain					0.0000
Bouyancy					-0.0001
Sum Static dp					0.4719
Net Fan VP	156.4	1,522.6	0.0698	14.70	0.1345
Fan Total Pressure					0.6064
Stack Exit	161.4	1,475.5	0.0698	14.70	0.1263

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 7I3DOL2 - GreenzoThermal Conditions

Tower Model = DPC 70
Power = 20.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,201.4 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %

Range = 32.40 °F
Altitude = 750.0 ft
Hot Water Temperature = 122.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 30.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 299.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.50 in
Plenum Hole Diameter = 10.25 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 86.7 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.133	Water Loading	5.15 gpm/ft ²
CF-1900SB - 5.00 ft	1.883	Pressure Ratio	15.09
Fill Total	1.883	L/G * KaV/L	2.1
Rain Zone	0.070	Fan:Box Ratio	36.7 %
Total KaV/L (CTI)	2.086	Fan Coverage	84.4 %
KaV/L Derate (%)	5	Airflow @ Fan	111,929.6 cfm/cell
HWT Correction (%)	6.6	Effective Fan Eff.	53.34 %
KaV/L Adjusted	1.844	Dry Air Rate	7,035.7 lb/min/cell
L/G Ratio	1.118	Total Fill Ht.	5 ft
		Fill Area	213.8 ft ² /cell
		Evaporation Rate	2.73 %
		Evaporation Rate	52.1 gpm (tower)
		Inlet DBT	89.3 °F
		Exit WBT	106.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	299.3	693.3	0.0694	14.74	0.0277
Rain Zone	427.5	485.3	0.0694	14.74	0.0134
Fill	427.5	504.5	0.0678	15.33	0.2816
Spray Zone	427.5	523.6	0.0662	15.33	0.0671
Drift Eliminator	427.5	523.6	0.0662	15.91	0.0282
Fan Inlet	159.9	1,399.6	0.0662	15.91	0.0539
Regain					0.0000
Bouyancy					-0.0042
Sum Static dp					0.4677
Net Fan VP	159.9	1,399.6	0.0662	15.91	0.1078
Fan Total Pressure					0.5755
Stack Exit	165.0	1,356.5	0.0662	15.91	0.1012

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 7H5DO2 - GreenzoThermal Conditions

Tower Model = DPC 70
Power = 20.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,201.4 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 32.40 °F
Altitude = 750.0 ft
Hot Water Temperature = 122.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 4.90 ft
Total Fill Height = 4.9 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.65 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 28.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 279.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.85 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 91.2 %

KaV/L Data

Spray Zone	0.135
CF-1200MABT - 4.90 ft	2.117
Fill Total	2.117
Rain Zone	0.068
Total KaV/L (CTI)	2.320
KaV/L Derate (%)	5
HWT Correction (%)	6.6
KaV/L Adjusted	2.050
L/G Ratio	1.284

Thermal Data - Misc.

Water Loading	5.91 gpm/ft ²
Pressure Ratio	23.68
L/G * KaV/L	2.6
Fan:Box Ratio	41.4 %
Fan Coverage	91.3 %
Airflow @ Fan	103,667.5 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	6,444.7 lb/min/cell
Total Fill Ht.	4.9 ft
Fill Area	186.2 ft ² /cell
Evaporation Rate	2.76 %
Evaporation Rate	55.4 gpm (tower)
Inlet DBT	91.0 °F
Exit WBT	108.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	279.3	682.1	0.0692	14.78	0.0267
Rain Zone	372.4	511.6	0.0692	14.78	0.0166
Fill	372.4	534.2	0.0675	15.43	0.4757
Spray Zone	372.4	556.8	0.0658	15.43	0.0821
Drift Eliminator	372.4	556.8	0.0658	16.09	0.0321
Fan Inlet	157.3	1,318.3	0.0658	16.09	0.0475
Regain					0.0000
Bouyancy					-0.0047
Sum Static dp					0.6760
Net Fan VP	157.3	1,318.3	0.0658	16.09	0.0950
Fan Total Pressure					0.7709
Stack Exit	162.4	1,277.0	0.0658	16.09	0.0891

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** ATP - 430 X 2Thermal Conditions

Tower Model = DPC 70
Power = 15.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,641.8 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 10.80 °F
Altitude = 750.0 ft
Hot Water Temperature = 100.40 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft
Total Fill Height = 6 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 28.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 279.3 ft²

Plenum & Fan Section

Drift Eliminator: CF150MAx
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.50 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 106.6 %

KaV/L Data

Spray Zone	0.119
CF-1900SB - 6.00 ft	1.439
Fill Total	1.439
Rain Zone	0.060
Total KaV/L (CTI)	1.619
KaV/L Derate (%)	5
HWT Correction (%)	0.1
KaV/L Adjusted	1.536
L/G Ratio	1.922

Thermal Data - Misc.

Water Loading	7.09 gpm/ft ²
Pressure Ratio	22.12
L/G * KaV/L	3.0
Fan:Box Ratio	41.4 %
Fan Coverage	90.3 %
Airflow @ Fan	93,387.7 cfm/cell
Effective Fan Eff.	62.87 %
Dry Air Rate	6,071.9 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	186.2 ft ² /cell
Evaporation Rate	0.95 %
Evaporation Rate	26.8 gpm (tower)
Inlet DBT	91.0 °F
Exit WBT	97.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	279.3	642.6	0.0692	14.78	0.0237
Rain Zone	372.4	482.0	0.0692	14.78	0.0215
Fill	372.4	491.8	0.0684	15.08	0.3461
Spray Zone	372.4	501.5	0.0677	15.08	0.0880
Drift Eliminator	372.4	501.5	0.0677	15.38	0.0458
Fan Inlet	157.3	1,187.6	0.0677	15.38	0.0396
Regain					0.0000
Bouyancy					-0.0021
Sum Static dp					0.5626
Net Fan VP	157.3	1,187.6	0.0677	15.38	0.0793
Fan Total Pressure					0.6419
Stack Exit	162.4	1,150.4	0.0677	15.38	0.0744

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 7G3DO2Thermal Conditions

Tower Model = DPC 70
Power = 12.50 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 1,673.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 18.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 107.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 12.00 ft
Cell Width = 12.00 ft
Net Total Inlet Area = 239.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 9.00 ft
Seal Disk/Hub Diameter = 1.62 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 9.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 97.1 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.143	Water Loading	6.11 gpm/ft ²
CF-1200MABT - 2.94 ft	1.445	Pressure Ratio	21.80
Fill Total	1.445	L/G * KaV/L	2.0
Rain Zone	0.068	Fan:Box Ratio	45.8 %
Total KaV/L (CTI)	1.655	Fan Coverage	96.5 %
KaV/L Derate (%)	5	Airflow @ Fan	78,898.2 cfm/cell
HWT Correction (%)	2.3	Effective Fan Eff.	61.72 %
KaV/L Adjusted	1.535	Dry Air Rate	5,095.3 lb/min/cell
L/G Ratio	1.318	Total Fill Ht.	2.94 ft
		Fill Area	136.8 ft ² /cell
		Evaporation Rate	1.57 %
		Evaporation Rate	25.5 gpm (tower)
		Inlet DBT	91.0 °F
		Exit WBT	99.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	239.4	629.2	0.0692	14.78	0.0228
Rain Zone	273.6	550.5	0.0692	14.78	0.0192
Fill	273.6	563.6	0.0683	15.13	0.3244
Spray Zone	273.6	576.7	0.0674	15.13	0.0942
Drift Eliminator	273.6	576.7	0.0674	15.48	0.0355
Fan Inlet	127.9	1,234.1	0.0674	15.48	0.0426
Regain					0.0000
Bouyancy					-0.0024
Sum Static dp					0.5363
Net Fan VP	127.9	1,234.1	0.0674	15.48	0.0852
Fan Total Pressure					0.6215
Stack Exit	132.0	1,195.5	0.0674	15.48	0.0800

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** IJL 100 with trickle, 4ft fan, 3 hpThermal Conditions

Tower Model = DPC 70
Power = 3.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 264.0 gpm
Wet Bulb Temperature = 82.94 °F
Relative Humidity = 70.00 %

Range = 9.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 98.60 °F
Approach = 6.66 °F

Fill Section

Spray Height = 1.25 ft
HTP20 = 4.43 ft
Total Fill Height = 4.43 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.18 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 7.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 93.1 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft

Fan Tip Clearance = 0.60 in
Plenum Hole Diameter = 4.10 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 1.82 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 106.7 %

KaV/L Data

Spray Zone	0.105
HTP20 - 4.43 ft	1.140
Fill Total	1.140
Rain Zone	0.061
Total KaV/L (CTI)	1.305
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.240
L/G Ratio	1.800

Thermal Data - Misc.

Water Loading	5.67 gpm/ft ²
Pressure Ratio	128.14
L/G * KaV/L	2.2
Fan:Box Ratio	26.9 %
Fan Coverage	89.3 %
Airflow @ Fan	19,765.3 cfm/cell
Effective Fan Eff.	53.34 %
Dry Air Rate	1,297.0 lb/min/cell
Total Fill Ht.	4.43 ft
Fill Area	46.6 ft ² /cell
Evaporation Rate	0.81 %
Evaporation Rate	2.3 gpm (tower)
Inlet DBT	91.6 °F
Exit WBT	95.2 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	93.1	206.3	0.0691	14.81	0.0024
Rain Zone	46.6	412.5	0.0691	14.81	0.0157
Fill	46.6	418.6	0.0686	15.02	0.2167
Spray Zone	46.6	424.6	0.0681	15.02	0.0601
Drift Eliminator	46.6	424.6	0.0681	15.24	0.0180
Fan Inlet	12.8	1,544.7	0.0681	15.24	0.0675
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3796
Net Fan VP	12.8	1,544.7	0.0681	15.24	0.1350
Fan Total Pressure					0.5145
Stack Exit	13.2	1,497.1	0.0681	15.24	0.1268

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 185 X 2Thermal Conditions

Tower Model = DPC 70
Power = 7.50 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 1,056.7 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 18.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 107.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 20.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 10.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 199.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 7.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 7.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 1.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 102.2 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.128	Water Loading 5.56 gpm/ft ²
CF-1900SB - 5.00 ft 1.571	Pressure Ratio 33.74
Fill Total 1.571	L/G * KaV/L 2.3
Rain Zone 0.066	Fan:Box Ratio 40.3 %
Total KaV/L (CTI) 1.765	Fan Coverage 86.1 %
KaV/L Derate (%) 5	Airflow @ Fan 49,062.2 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 56.79 %
KaV/L Adjusted 1.636	Dry Air Rate 3,156.1 lb/min/cell
L/G Ratio 1.416	Total Fill Ht. 5 ft
	Fill Area 95.0 ft ² /cell
	Evaporation Rate 1.56 %
	Evaporation Rate 16.9 gpm (tower)
	Inlet DBT 91.0 °F
	Exit WBT 100.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	199.5	467.6	0.0692	14.78	0.0126
Rain Zone	190.0	491.0	0.0692	14.78	0.0168
Fill	190.0	503.7	0.0682	15.16	0.2899
Spray Zone	190.0	516.4	0.0672	15.16	0.0770
Drift Eliminator	190.0	516.4	0.0672	15.55	0.0278
Fan Inlet	78.8	1,244.5	0.0672	15.55	0.0432
Regain					0.0000
Bouyancy					-0.0020
Sum Static dp					0.4653
Net Fan VP	78.8	1,244.5	0.0672	15.55	0.0865
Fan Total Pressure					0.5517
Stack Exit	80.7	1,216.3	0.0672	15.55	0.0826

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 410 X 2 GopaldasThermal Conditions

Tower Model = DPC 70
Power = 15.00 HP
Cold Water Temperature = 85.00 °F
Flow Rate = 2,400.0 gpm
Wet Bulb Temperature = 80.00 °F
Relative Humidity = 70.00 %

Range = 10.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 95.00 °F
Approach = 5.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 3.92 ft
Total Fill Height = 3.92 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 8.67 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 28.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 14.00 ft
Cell Width = 14.00 ft
Net Total Inlet Area = 279.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.17 ft
Fan Stack Height = 2.50 ft
Regain Height = N/A
Plenum Height = 2.83 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 90.1 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.130	Water Loading	6.44 gpm/ft ²
CF-1200MABT - 3.92 ft	1.688	Pressure Ratio	19.64
Fill Total	1.688	L/G * KaV/L	2.5
Rain Zone	0.066	Fan:Box Ratio	41.4 %
Total KaV/L (CTI)	1.884	Fan Coverage	91.1 %
KaV/L Derate (%)	5	Airflow @ Fan	95,390.7 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	62.87 %
KaV/L Adjusted	1.790	Dry Air Rate	6,341.2 lb/min/cell
L/G Ratio	1.414	Total Fill Ht.	3.92 ft
		Fill Area	186.2 ft ² /cell
		Evaporation Rate	0.90 %
		Evaporation Rate	19.5 gpm (tower)
		Inlet DBT	88.4 °F
		Exit WBT	91.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	279.3	666.0	0.0696	14.67	0.0256
Rain Zone	372.4	499.5	0.0696	14.67	0.0174
Fill	372.4	505.9	0.0692	14.86	0.3540
Spray Zone	372.4	512.3	0.0687	14.86	0.0787
Drift Eliminator	372.4	512.3	0.0687	15.04	0.0279
Fan Inlet	157.3	1,213.1	0.0687	15.04	0.0420
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.5444
Net Fan VP	157.3	1,213.1	0.0687	15.04	0.0840
Fan Total Pressure					0.6284
Stack Exit	162.4	1,175.1	0.0687	15.04	0.0788

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** New CTI Series DPC - 70
Delhi, Delhi
India**Project Description:** PFRP CTI Series**Comments:** 450X2 GopaldasThermal Conditions

Tower Model = DPC 70
Power = 20.00 HP
Cold Water Temperature = 85.00 °F
Flow Rate = 2,400.0 gpm
Wet Bulb Temperature = 80.00 °F
Relative Humidity = 75.00 %

Range = 10.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 95.00 °F
Approach = 5.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 30.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 15.00 ft
Cell Width = 15.00 ft
Net Total Inlet Area = 299.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 10.00 ft
Seal Disk/Hub Diameter = 1.80 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 10.25 ft
Fan Stack Height = 2.50 ft
Regain Height = N/A
Plenum Height = 2.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 101.5 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.142	Water Loading	5.61 gpm/ft ²
CF-1200MABT - 2.94 ft	1.479	Pressure Ratio	13.98
Fill Total	1.479	L/G * KaV/L	2.1
Rain Zone	0.068	Fan:Box Ratio	36.1 %
Total KaV/L (CTI)	1.689	Fan Coverage	85.2 %
KaV/L Derate (%)	5	Airflow @ Fan	118,297.8 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	62.87 %
KaV/L Adjusted	1.604	Dry Air Rate	7,890.3 lb/min/cell
L/G Ratio	1.280	Total Fill Ht.	2.94 ft
		Fill Area	213.8 ft ² /cell
		Evaporation Rate	0.88 %
		Evaporation Rate	21.5 gpm (tower)
		Inlet DBT	86.7 °F
		Exit WBT	90.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	299.3	771.6	0.0698	14.63	0.0345
Rain Zone	427.5	540.1	0.0698	14.63	0.0182
Fill	427.5	546.8	0.0693	14.81	0.3088
Spray Zone	427.5	553.4	0.0689	14.81	0.0880
Drift Eliminator	427.5	553.4	0.0689	14.99	0.0331
Fan Inlet	157.3	1,504.4	0.0689	14.99	0.0647
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					0.5461
Net Fan VP	157.3	1,504.4	0.0689	14.99	0.1295
Fan Total Pressure					0.6756
Stack Exit	162.4	1,457.2	0.0689	14.99	0.1215

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** HVAC Solutions Vinay Dutt Length - 48ft Width - 16ft Height - 11ft Fan Dia - 12ft Motor Power - 60HPThermal Conditions

Tower Model = DFC 70UX
Power = 60.95 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 8,900.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 3.50 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.71 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft

No. of Cells = 3
Inlet Height = 3.50 ft
Cell Length = 16.00 ft
Cell Width = 16.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 2.79 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 11.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 60.95 HP

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.182	Water Loading 12.2 gpm/ft ²
CF-1200MABT - 1.96 ft 0.792	Pressure Ratio 9.60
Fill Total 0.792	L/G * KaV/L 1.9
Rain Zone 0.064	Fan:Box Ratio 45.4 %
Total KaV/L (CTI) 1.039	Fan Coverage 89.6 %
KaV/L Derate (%) 5	Airflow @ Fan 197,711.7 cfm/cell
HWT Correction (%) 0.0	Effective Fan Eff. 65.40 %
KaV/L Adjusted 0.987	Dry Air Rate 13,035.5 lb/min/cell
L/G Ratio 1.887	Total Fill Ht. 1.96 ft
	Fill Area 243.2 ft ² /cell
	Evaporation Rate 0.69 %
	Evaporation Rate 61.5 gpm (tower)
	Inlet DBT 91.7 °F
	Exit WBT 93.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,360.8	0.0691	14.81	0.1063
Rain Zone	729.6	793.8	0.0691	14.81	0.0531
Fill	729.6	803.4	0.0687	14.99	0.5305
Spray Zone	729.6	813.0	0.0683	14.99	0.2566
Drift Eliminator	729.6	813.0	0.0683	15.17	0.0739
Fan Inlet	337.8	1,755.9	0.0683	15.17	0.0875
Regain					0.0000
Bouyancy					-0.0013
Sum Static dp					1.1066
Net Fan VP	337.8	1,755.9	0.0683	15.17	0.1750
Fan Total Pressure					1.2816
Stack Exit	348.8	1,700.6	0.0683	15.17	0.1641

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6233HUX (16 x 8 feet) - 3.97 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 5.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 815.3 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 16.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 8.00 ft
Cell Width = 8.00 ft
Net Total Inlet Area = 182.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft

Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 110.2 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.115	Water Loading	6.7 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	49.20
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	32.2 %
Total KaV/L (CTI)	0.998	Fan Coverage	98.0 %
KaV/L Derate (%)	5	Airflow @ Fan	31,104.6 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.948	Dry Air Rate	2,049.6 lb/min/cell
L/G Ratio	1.817	Total Fill Ht.	1.96 ft
		Fill Area	60.8 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	6.0 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	182.4	332.6	0.0692	14.80	0.0064
Rain Zone	121.6	498.9	0.0692	14.80	0.0253
Fill	121.6	505.3	0.0687	14.99	0.1812
Spray Zone	121.6	511.6	0.0683	14.99	0.0722
Drift Eliminator	121.6	511.6	0.0683	15.18	0.0276
Fan Inlet	40.0	1,555.8	0.0683	15.18	0.0686
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3805
Net Fan VP	40.0	1,555.8	0.0683	15.18	0.1372
Fan Total Pressure					0.5177
Stack Exit	41.3	1,507.8	0.0683	15.18	0.1289

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6173 TESTThermal Conditions

Tower Model = DFC 60HUX
Power = 2.32 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 615.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft
Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 2.32 HP

KaV/L Data

Spray Zone	0.106
CF-1200MABT - 1.96 ft	0.819
Fill Total	0.819
Rain Zone	0.067
Total KaV/L (CTI)	0.991
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	0.942
L/G Ratio	1.805

Thermal Data - Misc.

Water Loading	6.62 gpm/ft ²
Pressure Ratio	63.30
L/G * KaV/L	1.7
Fan:Box Ratio	42.1 %
Fan Coverage	100.0 %
Airflow @ Fan	21,474.4 cfm/cell
Effective Fan Eff.	53.34 %
Dry Air Rate	1,415.3 lb/min/cell
Total Fill Ht.	1.96 ft
Fill Area	46.6 ft ² /cell
Evaporation Rate	0.67 %
Evaporation Rate	4.1 gpm (tower)
Inlet DBT	90.0 °F
Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	262.5	0.0692	14.80	0.0040
Rain Zone	93.1	450.0	0.0692	14.80	0.0212
Fill	93.1	455.7	0.0687	14.99	0.1459
Spray Zone	93.1	461.3	0.0683	14.99	0.0576
Drift Eliminator	93.1	461.3	0.0683	15.17	0.0219
Fan Inlet	40.0	1,074.1	0.0683	15.17	0.0327
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.2825
Net Fan VP	40.0	1,074.1	0.0683	15.17	0.0654
Fan Total Pressure					0.3479
Stack Exit	41.3	1,041.0	0.0683	15.17	0.0614

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2200

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 3,632.5 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 32.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 124.9 %

KaV/L Data

Spray Zone	0.124
CF-1900SB - 7.00 ft	2.247
Fill Total	2.247
Rain Zone	0.068
Total KaV/L (CTI)	2.439
KaV/L Derate (%)	5
HWT Correction (%)	6.6
KaV/L Adjusted	2.156
L/G Ratio	1.239

Thermal Data - Misc.

Water Loading	3.73 gpm/ft ²
Pressure Ratio	8.13
L/G * KaV/L	2.7
Fan:Box Ratio	37.7 %
Fan Coverage	84.7 %
Airflow @ Fan	487,986.1 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	30,182.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	2.37 %
Evaporation Rate	107.4 gpm (tower)
Inlet DBT	99.0 °F
Exit WBT	110.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,076.4	0.0680	15.18	0.0654
Rain Zone	972.8	470.9	0.0680	15.18	0.0136
Fill	972.8	486.3	0.0668	15.67	0.3634
Spray Zone	972.8	501.6	0.0656	15.67	0.0638
Drift Eliminator	972.8	501.6	0.0656	16.17	0.0254
Fan Inlet	373.6	1,306.2	0.0656	16.17	0.0465
Regain					0.0000
Bouyancy					-0.0073
Sum Static dp					0.5709
Net Fan VP	373.6	1,306.2	0.0656	16.17	0.0930
Fan Total Pressure					0.6639
Stack Exit	385.9	1,264.5	0.0656	16.17	0.0871

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 850

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 1,541.0 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 113.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 3.00 ft
Total Fill Height = 3 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 22.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 22.00 ft
Cell Width = 22.00 ft
Net Total Inlet Area = 292.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.7 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.139	Water Loading 3.35 gpm/ft ²
CF-1900SB - 3.00 ft 1.445	Pressure Ratio 8.55
Fill Total 1.445	L/G * KaV/L 1.4
Rain Zone 0.074	Fan:Box Ratio 32.6 %
Total KaV/L (CTI) 1.659	Fan Coverage 86.9 %
KaV/L Derate (%) 5	Airflow @ Fan 239,637.7 cfm/cell
HWT Correction (%) 3.9	Effective Fan Eff. 67.58 %
KaV/L Adjusted 1.511	Dry Air Rate 15,367.2 lb/min/cell
L/G Ratio 0.901	Total Fill Ht. 3 ft
	Fill Area 459.8 ft ² /cell
	Evaporation Rate 1.66 %
	Evaporation Rate 27.8 gpm (tower)
	Inlet DBT 99.0 °F
	Exit WBT 101.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	292.6	797.1	0.0680	15.18	0.0359
Rain Zone	459.8	507.3	0.0680	15.18	0.0116
Fill	459.8	514.2	0.0675	15.39	0.1714
Spray Zone	459.8	521.2	0.0671	15.39	0.0597
Drift Eliminator	459.8	521.2	0.0671	15.59	0.0283
Fan Inlet	152.6	1,570.0	0.0671	15.59	0.0687
Regain					0.0000
Bouyancy					-0.0021
Sum Static dp					0.3735
Net Fan VP	152.6	1,570.0	0.0671	15.59	0.1374
Fan Total Pressure					0.5109
Stack Exit	157.6	1,520.3	0.0671	15.59	0.1288

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1750 X 4 Eriphase

Thermal Conditions

Tower Model = 70	Range = 25.20 °F
Power = 60.00 HP	Altitude = 800.0 ft
Cold Water Temperature = 89.60 °F	Hot Water Temperature = 114.80 °F
Flow Rate = 18,052.0 gpm	Approach = 8.64 °F
Wet Bulb Temperature = 80.96 °F	
Relative Humidity = 75.00 %	

Fill Section

Spray Height = 1.25 ft	Nozzle Type = DekSpray Nozzle
CF-1900SB = 7.00 ft	KaV/L Derate = 5.0 %
	dP Derate = 5.0 %
Total Fill Height = 7 ft	Fill Obstruction = 5.0 %
Rain Height = 5.00 ft	Total Height Fill + Rain + Spray = 13.25 ft

Tower Geometry

Cells: Back-to-back	No. of Cells = 4
End Walls: Both Ends Open	Inlet Height = 5.00 ft
Louver Type: No Louvers	Cell Length = 28.00 ft
Louver Coefficient: 1.00	Cell Width = 28.00 ft
Inlet Obstruction = 5.0 %	Net Total Inlet Area = 1,064.0 ft ²
Tower Length = 56.00 ft	

Plenum & Fan Section

Drift Eliminator: DE120	Fan Tip Clearance = 1.00 in
Drift Obstruction = 5.0 %	Plenum Hole Diameter = 20.17 ft
Fan Inlet Loss Coefficient = 0.50	Fan Stack Height = 6.00 ft
Transmission Efficiency = 95.0 %	Regain Height = N/A
Total Fan Efficiency = 75.0 %	Plenum Height = 5.75 ft
Fan Diameter = 20.00 ft	Spray to Drift Height = 1.50 ft
Seal Disk/Hub Diameter = 3.60 ft	Fan Deck Height = 20.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.8 %

KaV/L Data

Spray Zone	0.119
CF-1900SB - 7.00 ft	1.915
Fill Total	1.915
Rain Zone	0.082
Total KaV/L (CTI)	2.116
KaV/L Derate (%)	5
HWT Correction (%)	4.4
KaV/L Adjusted	1.916
L/G Ratio	1.536

Thermal Data - Misc.

Water Loading	6.06 gpm/ft ²
Pressure Ratio	6.13
L/G * KaV/L	2.9
Fan:Box Ratio	40.7 %
Fan Coverage	91.1 %
Airflow @ Fan	371,141.1 cfm/cell
Effective Fan Eff.	71.02 %
Dry Air Rate	23,243.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	2.11 %
Evaporation Rate	365.1 gpm (tower)
Inlet DBT	87.8 °F
Exit WBT	106.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,064.0	1,284.9	0.0695	14.70	0.0953
Rain Zone	2,979.2	458.9	0.0695	14.70	0.0250
Fill	2,979.2	478.6	0.0678	15.34	0.3661
Spray Zone	2,979.2	498.3	0.0660	15.34	0.0725
Drift Eliminator	2,979.2	498.3	0.0660	15.97	0.0252
Fan Inlet	1,237.0	1,200.2	0.0660	15.97	0.0395
Regain					0.0000
Bouyancy					-0.0093
Sum Static dp					0.6142
Net Fan VP	1,237.0	1,200.2	0.0660	15.97	0.0790
Fan Total Pressure					0.6933
Stack Exit	1,277.7	1,161.9	0.0660	15.97	0.0741

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 70x35ft Dimensions Fan Deck Height 18ft 4.5ft Air Inlet 5ft Layer Fill

Thermal Conditions

Tower Model = 70	
Power = 75.00 HP	Range = 18.00 °F
Cold Water Temperature = 89.60 °F	Altitude = 710.0 ft
Flow Rate = 14,530.0 gpm	Hot Water Temperature = 107.60 °F
Wet Bulb Temperature = 80.60 °F	Approach = 9.00 °F
Relative Humidity = 75.00 %	

Fill Section

Spray Height = 1.25 ft	Nozzle Type = DekSpray Nozzle
CF-1900SB = 5.00 ft	KaV/L Derate = 5.0 %
	dP Derate = 5.0 %
Total Fill Height = 5 ft	Fill Obstruction = 5.0 %
Rain Height = 4.50 ft	Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline	No. of Cells = 2
End Walls: Both Ends Open	Inlet Height = 4.50 ft
Louver Type: No Louvers	Cell Length = 35.00 ft
Louver Coefficient: 1.00	Cell Width = 35.00 ft
Inlet Obstruction = 5.0 %	Net Total Inlet Area = 897.8 ft ²
Tower Length = 70.00 ft	

Plenum & Fan Section

Drift Eliminator: DE120	Fan Tip Clearance = 1.00 in
Drift Obstruction = 5.0 %	Plenum Hole Diameter = 24.17 ft
Fan Inlet Loss Coefficient = 0.50	Fan Stack Height = 8.00 ft
Transmission Efficiency = 95.0 %	Regain Height = N/A
Total Fan Efficiency = 75.0 %	Plenum Height = 5.75 ft
Fan Diameter = 24.00 ft	Spray to Drift Height = 1.50 ft
Seal Disk/Hub Diameter = 4.32 ft	Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 94.6 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.118	Water Loading 6.24 gpm/ft ²
CF-1900SB - 5.00 ft 1.449	Pressure Ratio 5.62
Fill Total 1.449	L/G * KaV/L 2.4
Rain Zone 0.076	Fan:Box Ratio 37.4 %
Total KaV/L (CTI) 1.643	Fan Coverage 81.1 %
KaV/L Derate (%) 5	Airflow @ Fan 562,307.0 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 71.90 %
KaV/L Adjusted 1.524	Dry Air Rate 36,156.5 lb/min/cell
L/G Ratio 1.573	Total Fill Ht. 5 ft
	Fill Area 1,163.8 ft ² /cell
	Evaporation Rate 1.51 %
	Evaporation Rate 208.2 gpm (tower)
	Inlet DBT 87.4 °F
	Exit WBT 100.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	897.8	1,179.1	0.0698	14.64	0.0806
Rain Zone	2,327.5	454.8	0.0698	14.64	0.0222
Fill	2,327.5	469.0	0.0685	15.09	0.2547
Spray Zone	2,327.5	483.2	0.0672	15.09	0.0711
Drift Eliminator	2,327.5	483.2	0.0672	15.55	0.0239
Fan Inlet	888.1	1,266.4	0.0672	15.55	0.0448
Regain					0.0000
Bouyancy					-0.0078
Sum Static dp					0.4896
Net Fan VP	888.1	1,266.4	0.0672	15.55	0.0895
Fan Total Pressure					0.5791
Stack Exit	917.4	1,225.9	0.0672	15.55	0.0839

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Test MT

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 14,530.0 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 4.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 2
Inlet Height = 4.50 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 897.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 94.6 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.118	Water Loading 6.24 gpm/ft ²
CF-1900SB - 5.00 ft 1.449	Pressure Ratio 5.62
Fill Total 1.449	L/G * KaV/L 2.4
Rain Zone 0.076	Fan:Box Ratio 37.4 %
Total KaV/L (CTI) 1.643	Fan Coverage 81.1 %
KaV/L Derate (%) 5	Airflow @ Fan 562,307.0 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 71.90 %
KaV/L Adjusted 1.524	Dry Air Rate 36,156.5 lb/min/cell
L/G Ratio 1.573	Total Fill Ht. 5 ft
	Fill Area 1,163.8 ft ² /cell
	Evaporation Rate 1.51 %
	Evaporation Rate 208.2 gpm (tower)
	Inlet DBT 87.4 °F
	Exit WBT 100.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	897.8	1,179.1	0.0698	14.64	0.0806
Rain Zone	2,327.5	454.8	0.0698	14.64	0.0222
Fill	2,327.5	469.0	0.0685	15.09	0.2547
Spray Zone	2,327.5	483.2	0.0672	15.09	0.0711
Drift Eliminator	2,327.5	483.2	0.0672	15.55	0.0239
Fan Inlet	888.1	1,266.4	0.0672	15.55	0.0448
Regain					0.0000
Bouyancy					-0.0078
Sum Static dp					0.4896
Net Fan VP	888.1	1,266.4	0.0672	15.55	0.0895
Fan Total Pressure					0.5791
Stack Exit	917.4	1,225.9	0.0672	15.55	0.0839

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L3GOL1 3300 CMh, Mahathermal

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 14,530.0 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 13.25 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 4
Inlet Height = 5.00 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 912.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 96.3 %

KaV/L Data

Spray Zone	0.112
CF-1900SB - 7.00 ft	1.723
Fill Total	1.723
Rain Zone	0.079
Total KaV/L (CTI)	1.914
KaV/L Derate (%)	5
HWT Correction (%)	2.3
KaV/L Adjusted	1.775
L/G Ratio	1.768

Thermal Data - Misc.

Water Loading	6.64 gpm/ft ²
Pressure Ratio	8.12
L/G * KaV/L	3.1
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	256,517.1 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	16,367.9 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	1.51 %
Evaporation Rate	211.2 gpm (tower)
Inlet DBT	87.4 °F
Exit WBT	102.8 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	912.0	1,050.8	0.0698	14.64	0.0640
Rain Zone	2,188.8	437.9	0.0698	14.64	0.0263
Fill	2,188.8	453.3	0.0684	15.15	0.3362
Spray Zone	2,188.8	468.8	0.0669	15.15	0.0708
Drift Eliminator	2,188.8	468.8	0.0669	15.67	0.0222
Fan Inlet	795.0	1,290.6	0.0669	15.67	0.0463
Regain					0.0000
Bouyancy					-0.0072
Sum Static dp					0.5587
Net Fan VP	795.0	1,290.6	0.0669	15.67	0.0925
Fan Total Pressure					0.6513
Stack Exit	821.1	1,249.6	0.0669	15.67	0.0868

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Zydus 670

Thermal Conditions

Tower Model = 70
Power = 25.00 HP
Cold Water Temperature = 88.16 °F
Flow Rate = 2,130.0 gpm
Wet Bulb Temperature = 82.76 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 97.16 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 239.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 93.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.130	Water Loading	6.92 gpm/ft ²
CF-1200MABT - 2.94 ft	1.278	Pressure Ratio	17.70
Fill Total	1.278	L/G * KaV/L	2.2
Rain Zone	0.065	Fan:Box Ratio	35.9 %
Total KaV/L (CTI)	1.473	Fan Coverage	88.8 %
KaV/L Derate (%)	5	Airflow @ Fan	161,166.0 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	65.40 %
KaV/L Adjusted	1.399	Dry Air Rate	10,656.7 lb/min/cell
L/G Ratio	1.541	Total Fill Ht.	2.94 ft
		Fill Area	307.8 ft ² /cell
		Evaporation Rate	0.80 %
		Evaporation Rate	15.9 gpm (tower)
		Inlet DBT	89.7 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	239.4	656.1	0.0694	14.74	0.0248
Rain Zone	307.8	510.3	0.0694	14.74	0.0194
Fill	307.8	517.0	0.0690	14.93	0.2798
Spray Zone	307.8	523.6	0.0685	14.93	0.0864
Drift Eliminator	307.8	523.6	0.0685	15.12	0.0292
Fan Inlet	112.6	1,431.4	0.0685	15.12	0.0583
Regain					0.0000
Bouyancy					-0.0018
Sum Static dp					0.4960
Net Fan VP	112.6	1,431.4	0.0685	15.12	0.1165
Fan Total Pressure					0.6126
Stack Exit	116.3	1,386.2	0.0685	15.12	0.1093

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Mahathermal 3500CMH

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 4,402.3 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 16.20 °F
Altitude = 710.0 ft
Hot Water Temperature = 105.80 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 28.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 372.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 18.00 ft
Seal Disk/Hub Diameter = 3.24 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 18.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 100.2 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.121	Water Loading	5.91 gpm/ft ²
CF-1200MABT - 2.94 ft	1.281	Pressure Ratio	8.00
Fill Total	1.281	L/G * KaV/L	2.1
Rain Zone	0.064	Fan:Box Ratio	33.1 %
Total KaV/L (CTI)	1.467	Fan Coverage	85.8 %
KaV/L Derate (%)	5	Airflow @ Fan	366,732.7 cfm/cell
HWT Correction (%)	1.7	Effective Fan Eff.	70.27 %
KaV/L Adjusted	1.368	Dry Air Rate	23,762.2 lb/min/cell
L/G Ratio	1.537	Total Fill Ht.	2.94 ft
		Fill Area	744.8 ft ² /cell
		Evaporation Rate	1.37 %
		Evaporation Rate	60.4 gpm (tower)
		Inlet DBT	87.4 °F
		Exit WBT	98.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	372.4	934.0	0.0698	14.64	0.0506
Rain Zone	744.8	467.0	0.0698	14.64	0.0168
Fill	744.8	479.7	0.0687	15.04	0.2384
Spray Zone	744.8	492.4	0.0676	15.04	0.0737
Drift Eliminator	744.8	492.4	0.0676	15.43	0.0251
Fan Inlet	251.0	1,461.3	0.0676	15.43	0.0599
Regain					0.0000
Bouyancy					-0.0059
Sum Static dp					0.4586
Net Fan VP	251.0	1,461.3	0.0676	15.43	0.1198
Fan Total Pressure					0.5785
Stack Exit	259.2	1,414.8	0.0676	15.43	0.1123

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Kreston 4800

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 22,014.4 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 116.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 7.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 16.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 4
Inlet Height = 7.50 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 1,995.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 24.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 92.6 %

KaV/L Data

Spray Zone	0.133
HTP-25 - 7.84 ft	1.618
Fill Total	1.618
Rain Zone	0.121
Total KaV/L (CTI)	1.873
KaV/L Derate (%)	5
HWT Correction (%)	5.0
KaV/L Adjusted	1.686
L/G Ratio	1.095

Thermal Data - Misc.

Water Loading	4.73 gpm/ft ²
Pressure Ratio	5.33
L/G * KaV/L	1.8
Fan:Box Ratio	37.4 %
Fan Coverage	82.1 %
Airflow @ Fan	601,867.4 cfm/cell
Effective Fan Eff.	71.90 %
Dry Air Rate	38,428.3 lb/min/cell
Total Fill Ht.	7.84 ft
Fill Area	1,163.8 ft ² /cell
Evaporation Rate	2.29 %
Evaporation Rate	466.0 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	102.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,995.0	1,134.3	0.0695	14.72	0.0743
Rain Zone	4,655.0	486.1	0.0695	14.72	0.0328
Fill	4,655.0	501.7	0.0682	15.19	0.1952
Spray Zone	4,655.0	517.2	0.0669	15.19	0.0657
Drift Eliminator	4,655.0	517.2	0.0669	15.66	0.0277
Fan Inlet	1,776.1	1,355.4	0.0669	15.66	0.0511
Regain					0.0000
Bouyancy					-0.0079
Sum Static dp					0.4389
Net Fan VP	1,776.1	1,355.4	0.0669	15.66	0.1021
Fan Total Pressure					0.5410
Stack Exit	1,834.8	1,312.1	0.0669	15.66	0.0957

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7R5GO4

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 10,073.8 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 4.90 ft
Total Fill Height = 4.9 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.15 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 40.00 ft
No. of Cells = 4
Inlet Height = 4.00 ft
Cell Length = 20.00 ft
Cell Width = 20.00 ft
Net Total Inlet Area = 608.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.85 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 17.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 103.3 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	6.63 gpm/ft ²
CF-1200MABT - 4.90 ft	1.545	Pressure Ratio	7.67
Fill Total	1.545	L/G * KaV/L	3.0
Rain Zone	0.067	Fan:Box Ratio	39.4 %
Total KaV/L (CTI)	1.726	Fan Coverage	98.5 %
KaV/L Derate (%)	5	Airflow @ Fan	178,715.0 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	67.58 %
KaV/L Adjusted	1.640	Dry Air Rate	11,694.3 lb/min/cell
L/G Ratio	1.843	Total Fill Ht.	4.9 ft
		Fill Area	380.0 ft ² /cell
		Evaporation Rate	0.80 %
		Evaporation Rate	82.7 gpm (tower)
		Inlet DBT	91.3 °F
		Exit WBT	96.3 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	608.0	1,139.3	0.0692	14.81	0.0746
Rain Zone	1,520.0	455.7	0.0692	14.81	0.0221
Fill	1,520.0	463.0	0.0686	15.05	0.3772
Spray Zone	1,520.0	470.3	0.0680	15.05	0.0754
Drift Eliminator	1,520.0	470.3	0.0680	15.28	0.0228
Fan Inlet	610.5	1,170.8	0.0680	15.28	0.0387
Regain					0.0000
Bouyancy					-0.0032
Sum Static dp					0.6076
Net Fan VP	610.5	1,170.8	0.0680	15.28	0.0774
Fan Total Pressure					0.6850
Stack Exit	630.5	1,133.8	0.0680	15.28	0.0726

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6173HUX (14 x 7 feet) - 2.32 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 615.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft
Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 109.4 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	6.62 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	63.81
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	42.1 %
Total KaV/L (CTI)	0.996	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	23,351.6 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.947	Dry Air Rate	1,538.8 lb/min/cell
L/G Ratio	1.815	Total Fill Ht.	1.96 ft
		Fill Area	46.6 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.5 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	285.4	0.0692	14.80	0.0047
Rain Zone	93.1	489.3	0.0692	14.80	0.0245
Fill	93.1	495.5	0.0687	14.99	0.1738
Spray Zone	93.1	501.6	0.0683	14.99	0.0691
Drift Eliminator	93.1	501.6	0.0683	15.18	0.0264
Fan Inlet	40.0	1,168.0	0.0683	15.18	0.0387
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3364
Net Fan VP	40.0	1,168.0	0.0683	15.18	0.0773
Fan Total Pressure					0.4137
Stack Exit	41.3	1,132.0	0.0683	15.18	0.0727

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: DFC 60HUX Models (R5)
Delhi, Delhi
India

Project Description: Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height

Comments: 6303HUX (18 x 10 feet) - 6.52 BHP

Thermal Conditions

Tower Model = DFC 60HUX
Power = 7.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,078.7 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 105.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.106	Water Loading	6.31 gpm/ft ²
CF-1200MABT - 1.96 ft	0.819	Pressure Ratio	33.83
Fill Total	0.819	L/G * KaV/L	1.7
Rain Zone	0.067	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.991	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	39,465.3 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.942	Dry Air Rate	2,601.0 lb/min/cell
L/G Ratio	1.805	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	7.6 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	361.8	0.0692	14.80	0.0075
Rain Zone	171.0	450.3	0.0692	14.80	0.0212
Fill	171.0	455.9	0.0687	14.99	0.1461
Spray Zone	171.0	461.6	0.0683	14.99	0.0577
Drift Eliminator	171.0	461.6	0.0683	15.17	0.0219
Fan Inlet	57.9	1,363.2	0.0683	15.17	0.0527
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3062
Net Fan VP	57.9	1,363.2	0.0683	15.17	0.1054
Fan Total Pressure					0.4116
Stack Exit	59.7	1,321.4	0.0683	15.17	0.0990

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7M1GOL2

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 91.40 °F
Flow Rate = 8,805.8 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 16.20 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 6.86 ft
Total Fill Height = 6.86 ft
Rain Height = 4.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 12.61 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 56.00 ft
No. of Cells = 2
Inlet Height = 4.50 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 718.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 18.00 ft
Seal Disk/Hub Diameter = 3.24 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 18.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.89 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 102.4 %

KaV/L Data

Spray Zone	0.125
HTP-25 - 6.86 ft	1.134
Fill Total	1.134
Rain Zone	0.077
Total KaV/L (CTI)	1.336
KaV/L Derate (%)	5
HWT Correction (%)	2.3
KaV/L Adjusted	1.239
L/G Ratio	1.518

Thermal Data - Misc.

Water Loading	5.91 gpm/ft ²
Pressure Ratio	6.31
L/G * KaV/L	1.9
Fan:Box Ratio	33.1 %
Fan Coverage	86.3 %
Airflow @ Fan	380,690.5 cfm/cell
Effective Fan Eff.	70.27 %
Dry Air Rate	24,572.2 lb/min/cell
Total Fill Ht.	6.86 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	1.38 %
Evaporation Rate	124.7 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	99.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	718.2	1,007.4	0.0695	14.72	0.0586
Rain Zone	1,489.6	485.7	0.0695	14.72	0.0238
Fill	1,489.6	498.4	0.0684	15.11	0.1811
Spray Zone	1,489.6	511.1	0.0674	15.11	0.0789
Drift Eliminator	1,489.6	511.1	0.0674	15.49	0.0272
Fan Inlet	501.9	1,516.9	0.0674	15.49	0.0644
Regain					0.0000
Bouyancy					-0.0055
Sum Static dp					0.4285
Net Fan VP	501.9	1,516.9	0.0674	15.49	0.1288
Fan Total Pressure					0.5573
Stack Exit	518.4	1,468.7	0.0674	15.49	0.1207

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: DFC 60HUX Models (R5)
Delhi, Delhi
India

Project Description: Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height

Comments: 6123HUX (12x6 feet) - 2.09 BHP

Thermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 434.4 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 12.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 6.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 136.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft
Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.12 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.4 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.113	Water Loading	6.35 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	86.44
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	36.8 %
Total KaV/L (CTI)	0.996	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	17,082.2 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	41.88 %
KaV/L Adjusted	0.946	Dry Air Rate	1,125.6 lb/min/cell
L/G Ratio	1.815	Total Fill Ht.	1.96 ft
		Fill Area	34.2 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	3.3 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	136.8	243.6	0.0692	14.80	0.0034
Rain Zone	68.4	487.2	0.0692	14.80	0.0243
Fill	68.4	493.3	0.0687	14.99	0.1723
Spray Zone	68.4	499.5	0.0683	14.99	0.0685
Drift Eliminator	68.4	499.5	0.0683	15.18	0.0262
Fan Inlet	25.7	1,329.4	0.0683	15.18	0.0501
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3439
Net Fan VP	25.7	1,329.4	0.0683	15.18	0.1002
Fan Total Pressure					0.4441
Stack Exit	26.5	1,288.6	0.0683	15.18	0.0941

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6266HUX (16 x 9 feet) - 5.84 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 7.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 947.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 16.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 8.00 ft
Cell Width = 9.00 ft
Net Total Inlet Area = 190.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 109.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.117	Water Loading	6.92 gpm/ft ²
CF-1200MABT - 1.96 ft	0.814	Pressure Ratio	42.42
Fill Total	0.814	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	41.5 %
Total KaV/L (CTI)	0.999	Fan Coverage	99.1 %
KaV/L Derate (%)	5	Airflow @ Fan	35,681.9 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.949	Dry Air Rate	2,351.1 lb/min/cell
L/G Ratio	1.820	Total Fill Ht.	1.96 ft
		Fill Area	68.4 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	6.9 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	190.0	366.3	0.0692	14.80	0.0077
Rain Zone	136.8	508.8	0.0692	14.80	0.0262
Fill	136.8	515.2	0.0687	14.99	0.1889
Spray Zone	136.8	521.7	0.0683	14.99	0.0753
Drift Eliminator	136.8	521.7	0.0683	15.18	0.0288
Fan Inlet	57.9	1,232.5	0.0683	15.18	0.0431
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.3691
Net Fan VP	57.9	1,232.5	0.0683	15.18	0.0861
Fan Total Pressure					0.4552
Stack Exit	59.7	1,194.7	0.0683	15.18	0.0809

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L4GO1

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 3,750.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 10.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.00 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 4.90 ft
Total Fill Height = 4.9 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.65 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 319.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.85 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 16.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 107.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.111	Water Loading	6.85 gpm/ft ²
CF-1200MABT - 4.90 ft	1.435	Pressure Ratio	15.44
Fill Total	1.435	L/G * KaV/L	3.1
Rain Zone	0.059	Fan:Box Ratio	35.6 %
Total KaV/L (CTI)	1.605	Fan Coverage	87.0 %
KaV/L Derate (%)	5	Airflow @ Fan	254,489.9 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	69.17 %
KaV/L Adjusted	1.525	Dry Air Rate	16,571.7 lb/min/cell
L/G Ratio	2.006	Total Fill Ht.	4.9 ft
		Fill Area	547.2 ft ² /cell
		Evaporation Rate	0.87 %
		Evaporation Rate	34.7 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	97.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	319.2	765.8	0.0694	14.75	0.0338
Rain Zone	547.2	446.7	0.0694	14.75	0.0197
Fill	547.2	455.9	0.0686	15.05	0.3693
Spray Zone	547.2	465.1	0.0678	15.05	0.0770
Drift Eliminator	547.2	465.1	0.0678	15.36	0.0221
Fan Inlet	198.8	1,280.4	0.0678	15.36	0.0462
Regain					0.0000
Bouyancy					-0.0040
Sum Static dp					0.5642
Net Fan VP	198.8	1,280.4	0.0678	15.36	0.0923
Fan Total Pressure					0.6565
Stack Exit	205.3	1,239.8	0.0678	15.36	0.0865

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L3GOL4 - TKT

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 85.00 °F
Flow Rate = 11,007.0 gpm
Wet Bulb Temperature = 77.00 °F
Relative Humidity = 75.00 %
Range = 20.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 105.00 °F
Approach = 8.00 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 5.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 14.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 4
Inlet Height = 5.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 1,003.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 21.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.0 %

KaV/L Data

Spray Zone	0.137
HTP-25 - 7.84 ft	1.564
Fill Total	1.564
Rain Zone	0.096
Total KaV/L (CTI)	1.798
KaV/L Derate (%)	5
HWT Correction (%)	1.5
KaV/L Adjusted	1.681
L/G Ratio	1.144

Thermal Data - Misc.

Water Loading	5.03 gpm/ft ²
Pressure Ratio	5.83
L/G * KaV/L	1.9
Fan:Box Ratio	35.6 %
Fan Coverage	87.5 %
Airflow @ Fan	288,140.4 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	18,921.9 lb/min/cell
Total Fill Ht.	7.84 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	1.66 %
Evaporation Rate	173.6 gpm (tower)
Inlet DBT	83.5 °F
Exit WBT	95.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,003.2	1,092.1	0.0704	14.47	0.0697
Rain Zone	2,188.8	500.5	0.0704	14.47	0.0251
Fill	2,188.8	513.6	0.0693	14.85	0.2102
Spray Zone	2,188.8	526.6	0.0682	14.85	0.0725
Drift Eliminator	2,188.8	526.6	0.0682	15.23	0.0294
Fan Inlet	795.0	1,449.7	0.0682	15.23	0.0595
Regain					0.0000
Bouyancy					-0.0056
Sum Static dp					0.4608
Net Fan VP	795.0	1,449.7	0.0682	15.23	0.1190
Fan Total Pressure					0.5798
Stack Exit	821.1	1,403.7	0.0682	15.23	0.1116

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1700 TR 39 X 23

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 89.96 °F
Flow Rate = 20,412.0 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 8.44 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.40 °F
Approach = 5.76 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 4.00 ft
Total Fill Height = 4 ft
Rain Height = 5.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 92.00 ft
No. of Cells = 4
Inlet Height = 5.50 ft
Cell Length = 23.00 ft
Cell Width = 39.00 ft
Net Total Inlet Area = 1,369.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 20.00 ft
Seal Disk/Hub Diameter = 3.60 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 20.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 6.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.3 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 4.00 ft	1.120
Fill Total	1.120
Rain Zone	0.086
Total KaV/L (CTI)	1.327
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.260
L/G Ratio	1.722

Thermal Data - Misc.

Water Loading	5.99 gpm/ft ²
Pressure Ratio	5.41
L/G * KaV/L	2.2
Fan:Box Ratio	35.6 %
Fan Coverage	79.0 %
Airflow @ Fan	423,318.2 cfm/cell
Effective Fan Eff.	71.02 %
Dry Air Rate	27,834.6 lb/min/cell
Total Fill Ht.	4 ft
Fill Area	852.2 ft ² /cell
Evaporation Rate	0.76 %
Evaporation Rate	174.8 gpm (tower)
Inlet DBT	91.3 °F
Exit WBT	95.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,369.0	1,204.4	0.0692	14.81	0.0834
Rain Zone	3,408.6	483.7	0.0692	14.81	0.0335
Fill	3,408.6	490.2	0.0687	15.01	0.2267
Spray Zone	3,408.6	496.8	0.0682	15.01	0.0818
Drift Eliminator	3,408.6	496.8	0.0682	15.21	0.0259
Fan Inlet	1,237.0	1,368.9	0.0682	15.21	0.0531
Regain					0.0000
Bouyancy					-0.0027
Sum Static dp					0.5016
Net Fan VP	1,237.0	1,368.9	0.0682	15.21	0.1062
Fan Total Pressure					0.6078
Stack Exit	1,277.7	1,325.3	0.0682	15.21	0.0995

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2450 X 2

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 7,265.0 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
ShockWave 25 = 5.91 ft
Total Fill Height = 5.91 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 12.16 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 2
Inlet Height = 5.00 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 997.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.84 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 139.4 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.130	Water Loading	3.12 gpm/ft ²
ShockWave 25 - 5.91 ft	2.038	Pressure Ratio	5.43
Fill Total	2.038	L/G * KaV/L	2.2
Rain Zone	0.090	Fan:Box Ratio	37.4 %
Total KaV/L (CTI)	2.258	Fan Coverage	81.7 %
KaV/L Derate (%)	5	Airflow @ Fan	601,182.3 cfm/cell
HWT Correction (%)	6.6	Effective Fan Eff.	71.90 %
KaV/L Adjusted	1.996	Dry Air Rate	37,435.0 lb/min/cell
L/G Ratio	1.116	Total Fill Ht.	5.91 ft
		Fill Area	1,163.8 ft ² /cell
		Evaporation Rate	2.38 %
		Evaporation Rate	240.7 gpm (tower)
		Inlet DBT	99.0 °F
		Exit WBT	108.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	997.5	1,139.2	0.0680	15.18	0.0733
Rain Zone	2,327.5	488.2	0.0680	15.18	0.0202
Fill	2,327.5	502.4	0.0669	15.62	0.2126
Spray Zone	2,327.5	516.6	0.0659	15.62	0.0643
Drift Eliminator	2,327.5	516.6	0.0659	16.06	0.0272
Fan Inlet	888.1	1,353.9	0.0659	16.06	0.0502
Regain					0.0000
Bouyancy					-0.0065
Sum Static dp					0.4413
Net Fan VP	888.1	1,353.9	0.0659	16.06	0.1003
Fan Total Pressure					0.5417
Stack Exit	917.4	1,310.6	0.0659	16.06	0.0940

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2050

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 3,632.4 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Fill Height = 6 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 32.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.5 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 6.00 ft	2.120
Fill Total	2.120
Rain Zone	0.069
Total KaV/L (CTI)	2.310
KaV/L Derate (%)	5
HWT Correction (%)	6.6
KaV/L Adjusted	2.042
L/G Ratio	1.153

Thermal Data - Misc.

Water Loading	3.73 gpm/ft ²
Pressure Ratio	7.22
L/G * KaV/L	2.4
Fan:Box Ratio	37.7 %
Fan Coverage	84.7 %
Airflow @ Fan	466,159.2 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	28,967.6 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	2.37 %
Evaporation Rate	96.1 gpm (tower)
Inlet DBT	99.0 °F
Exit WBT	109.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,033.0	0.0680	15.18	0.0602
Rain Zone	972.8	452.0	0.0680	15.18	0.0119
Fill	972.8	465.6	0.0669	15.64	0.2841
Spray Zone	972.8	479.2	0.0658	15.64	0.0557
Drift Eliminator	972.8	479.2	0.0658	16.09	0.0230
Fan Inlet	373.6	1,247.8	0.0658	16.09	0.0425
Regain					0.0000
Bouyancy					-0.0067
Sum Static dp					0.4709
Net Fan VP	373.6	1,247.8	0.0658	16.09	0.0851
Fan Total Pressure					0.5560
Stack Exit	385.9	1,207.9	0.0658	16.09	0.0797

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6323HUXThermal Conditions

Tower Model = DFC 60HUX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,497.0 gpm
Wet Bulb Temperature = 82.94 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.06 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 84.1 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	8.75 gpm/ft ²
CF-1200MABT - 1.96 ft	0.811	Pressure Ratio	34.20
Fill Total	0.811	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.993	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	43,343.1 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.944	Dry Air Rate	2,856.0 lb/min/cell
L/G Ratio	1.827	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	8.5 gpm (tower)
		Inlet DBT	89.9 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	397.2	0.0692	14.80	0.0091
Rain Zone	171.0	494.3	0.0692	14.80	0.0250
Fill	171.0	500.6	0.0687	14.99	0.1779
Spray Zone	171.0	506.9	0.0683	14.99	0.0710
Drift Eliminator	171.0	506.9	0.0683	15.18	0.0271
Fan Inlet	57.9	1,497.1	0.0683	15.18	0.0635
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.3726
Net Fan VP	57.9	1,497.1	0.0683	15.18	0.1271
Fan Total Pressure					0.4997
Stack Exit	59.7	1,451.2	0.0683	15.18	0.1194

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Kreston 4300 X 2

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 19,813.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 116.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 7.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 16.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 64.00 ft
No. of Cells = 4
Inlet Height = 7.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 1,824.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 24.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 92.1 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.141	Water Loading 5.09 gpm/ft ²
HTP-25 - 7.84 ft 1.614	Pressure Ratio 6.19
Fill Total 1.614	L/G * KaV/L 1.9
Rain Zone 0.122	Fan:Box Ratio 37.7 %
Total KaV/L (CTI) 1.877	Fan Coverage 85.6 %
KaV/L Derate (%) 5	Airflow @ Fan 537,323.4 cfm/cell
HWT Correction (%) 5.0	Effective Fan Eff. 71.54 %
KaV/L Adjusted 1.689	Dry Air Rate 34,299.7 lb/min/cell
L/G Ratio 1.099	Total Fill Ht. 7.84 ft
	Fill Area 972.8 ft ² /cell
	Evaporation Rate 2.29 %
	Evaporation Rate 417.3 gpm (tower)
	Inlet DBT 89.3 °F
	Exit WBT 102.7 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,824.0	1,107.4	0.0695	14.72	0.0708
Rain Zone	3,891.2	519.1	0.0695	14.72	0.0368
Fill	3,891.2	535.7	0.0682	15.19	0.2229
Spray Zone	3,891.2	552.3	0.0669	15.19	0.0759
Drift Eliminator	3,891.2	552.3	0.0669	15.67	0.0321
Fan Inlet	1,494.4	1,438.2	0.0669	15.67	0.0575
Regain					0.0000
Bouyancy					-0.0079
Sum Static dp					0.4880
Net Fan VP	1,494.4	1,438.2	0.0669	15.67	0.1150
Fan Total Pressure					0.6029
Stack Exit	1,543.7	1,392.3	0.0669	15.67	0.1077

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6233HUX (16 x 8 feet) - 3.97 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 5.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 815.3 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 16.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 8.00 ft
Cell Width = 8.00 ft
Net Total Inlet Area = 182.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft

Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 110.2 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.115	Water Loading	6.7 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	49.20
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	32.2 %
Total KaV/L (CTI)	0.998	Fan Coverage	98.0 %
KaV/L Derate (%)	5	Airflow @ Fan	31,104.6 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.948	Dry Air Rate	2,049.6 lb/min/cell
L/G Ratio	1.817	Total Fill Ht.	1.96 ft
		Fill Area	60.8 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	6.0 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	182.4	332.6	0.0692	14.80	0.0064
Rain Zone	121.6	498.9	0.0692	14.80	0.0253
Fill	121.6	505.3	0.0687	14.99	0.1812
Spray Zone	121.6	511.6	0.0683	14.99	0.0722
Drift Eliminator	121.6	511.6	0.0683	15.18	0.0276
Fan Inlet	40.0	1,555.8	0.0683	15.18	0.0686
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3805
Net Fan VP	40.0	1,555.8	0.0683	15.18	0.1372
Fan Total Pressure					0.5177
Stack Exit	41.3	1,507.8	0.0683	15.18	0.1289

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6173 TESTThermal Conditions

Tower Model = DFC 60HUX
Power = 2.32 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 615.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft

Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 2.32 HP

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.106	Water Loading	6.62 gpm/ft ²
CF-1200MABT - 1.96 ft	0.819	Pressure Ratio	63.30
Fill Total	0.819	L/G * KaV/L	1.7
Rain Zone	0.067	Fan:Box Ratio	42.1 %
Total KaV/L (CTI)	0.991	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	21,474.4 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.942	Dry Air Rate	1,415.3 lb/min/cell
L/G Ratio	1.805	Total Fill Ht.	1.96 ft
		Fill Area	46.6 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.1 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	262.5	0.0692	14.80	0.0040
Rain Zone	93.1	450.0	0.0692	14.80	0.0212
Fill	93.1	455.7	0.0687	14.99	0.1459
Spray Zone	93.1	461.3	0.0683	14.99	0.0576
Drift Eliminator	93.1	461.3	0.0683	15.17	0.0219
Fan Inlet	40.0	1,074.1	0.0683	15.17	0.0327
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.2825
Net Fan VP	40.0	1,074.1	0.0683	15.17	0.0654
Fan Total Pressure					0.3479
Stack Exit	41.3	1,041.0	0.0683	15.17	0.0614

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2200

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 3,632.5 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Fill Height = 7 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 32.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 124.9 %

KaV/L Data

Spray Zone	0.124
CF-1900SB - 7.00 ft	2.247
Fill Total	2.247
Rain Zone	0.068
Total KaV/L (CTI)	2.439
KaV/L Derate (%)	5
HWT Correction (%)	6.6
KaV/L Adjusted	2.156
L/G Ratio	1.239

Thermal Data - Misc.

Water Loading	3.73 gpm/ft ²
Pressure Ratio	8.13
L/G * KaV/L	2.7
Fan:Box Ratio	37.7 %
Fan Coverage	84.7 %
Airflow @ Fan	487,986.1 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	30,182.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	2.37 %
Evaporation Rate	107.4 gpm (tower)
Inlet DBT	99.0 °F
Exit WBT	110.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,076.4	0.0680	15.18	0.0654
Rain Zone	972.8	470.9	0.0680	15.18	0.0136
Fill	972.8	486.3	0.0668	15.67	0.3634
Spray Zone	972.8	501.6	0.0656	15.67	0.0638
Drift Eliminator	972.8	501.6	0.0656	16.17	0.0254
Fan Inlet	373.6	1,306.2	0.0656	16.17	0.0465
Regain					0.0000
Bouyancy					-0.0073
Sum Static dp					0.5709
Net Fan VP	373.6	1,306.2	0.0656	16.17	0.0930
Fan Total Pressure					0.6639
Stack Exit	385.9	1,264.5	0.0656	16.17	0.0871

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 850

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 1,541.0 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 113.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 3.00 ft
Total Fill Height = 3 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 22.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 22.00 ft
Cell Width = 22.00 ft
Net Total Inlet Area = 292.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.7 %

KaV/L Data

Spray Zone	0.139
CF-1900SB - 3.00 ft	1.445
Fill Total	1.445
Rain Zone	0.074
Total KaV/L (CTI)	1.659
KaV/L Derate (%)	5
HWT Correction (%)	3.9
KaV/L Adjusted	1.511
L/G Ratio	0.901

Thermal Data - Misc.

Water Loading	3.35 gpm/ft ²
Pressure Ratio	8.55
L/G * KaV/L	1.4
Fan:Box Ratio	32.6 %
Fan Coverage	86.9 %
Airflow @ Fan	239,637.7 cfm/cell
Effective Fan Eff.	67.58 %
Dry Air Rate	15,367.2 lb/min/cell
Total Fill Ht.	3 ft
Fill Area	459.8 ft ² /cell
Evaporation Rate	1.66 %
Evaporation Rate	27.8 gpm (tower)
Inlet DBT	99.0 °F
Exit WBT	101.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	292.6	797.1	0.0680	15.18	0.0359
Rain Zone	459.8	507.3	0.0680	15.18	0.0116
Fill	459.8	514.2	0.0675	15.39	0.1714
Spray Zone	459.8	521.2	0.0671	15.39	0.0597
Drift Eliminator	459.8	521.2	0.0671	15.59	0.0283
Fan Inlet	152.6	1,570.0	0.0671	15.59	0.0687
Regain					0.0000
Bouyancy					-0.0021
Sum Static dp					0.3735
Net Fan VP	152.6	1,570.0	0.0671	15.59	0.1374
Fan Total Pressure					0.5109
Stack Exit	157.6	1,520.3	0.0671	15.59	0.1288

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1750 X 4 Eriphase

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 18,052.0 gpm
Wet Bulb Temperature = 80.96 °F
Relative Humidity = 75.00 %
Range = 25.20 °F
Altitude = 800.0 ft
Hot Water Temperature = 114.80 °F
Approach = 8.64 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 13.25 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 56.00 ft
No. of Cells = 4
Inlet Height = 5.00 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 1,064.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 20.00 ft
Seal Disk/Hub Diameter = 3.60 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 20.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.8 %

KaV/L Data

Spray Zone	0.119
CF-1900SB - 7.00 ft	1.915
Fill Total	1.915
Rain Zone	0.082
Total KaV/L (CTI)	2.116
KaV/L Derate (%)	5
HWT Correction (%)	4.4
KaV/L Adjusted	1.916
L/G Ratio	1.536

Thermal Data - Misc.

Water Loading	6.06 gpm/ft ²
Pressure Ratio	6.13
L/G * KaV/L	2.9
Fan:Box Ratio	40.7 %
Fan Coverage	91.1 %
Airflow @ Fan	371,141.1 cfm/cell
Effective Fan Eff.	71.02 %
Dry Air Rate	23,243.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	2.11 %
Evaporation Rate	365.1 gpm (tower)
Inlet DBT	87.8 °F
Exit WBT	106.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,064.0	1,284.9	0.0695	14.70	0.0953
Rain Zone	2,979.2	458.9	0.0695	14.70	0.0250
Fill	2,979.2	478.6	0.0678	15.34	0.3661
Spray Zone	2,979.2	498.3	0.0660	15.34	0.0725
Drift Eliminator	2,979.2	498.3	0.0660	15.97	0.0252
Fan Inlet	1,237.0	1,200.2	0.0660	15.97	0.0395
Regain					0.0000
Bouyancy					-0.0093
Sum Static dp					0.6142
Net Fan VP	1,237.0	1,200.2	0.0660	15.97	0.0790
Fan Total Pressure					0.6933
Stack Exit	1,277.7	1,161.9	0.0660	15.97	0.0741

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: DFC 60HUX Models (R5)
Delhi, Delhi
India

Project Description: Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height

Comments: 6206HUX (14 x 7 feet) - 3.84 BHP

Thermal Conditions

Tower Model = DFC 60HUX
Power = 5.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 733.4 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft
Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 109.5 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.130	Water Loading	7.88 gpm/ft ²
CF-1200MABT - 1.96 ft	0.808	Pressure Ratio	65.41
Fill Total	0.808	L/G * KaV/L	1.8
Rain Zone	0.069	Fan:Box Ratio	42.1 %
Total KaV/L (CTI)	1.007	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	27,522.0 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.957	Dry Air Rate	1,812.9 lb/min/cell
L/G Ratio	1.836	Total Fill Ht.	1.96 ft
		Fill Area	46.6 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	5.4 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	336.3	0.0692	14.80	0.0065
Rain Zone	93.1	576.4	0.0692	14.80	0.0329
Fill	93.1	583.8	0.0687	14.99	0.2483
Spray Zone	93.1	591.2	0.0682	14.99	0.0993
Drift Eliminator	93.1	591.2	0.0682	15.18	0.0379
Fan Inlet	40.0	1,376.6	0.0682	15.18	0.0537
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.4777
Net Fan VP	40.0	1,376.6	0.0682	15.18	0.1074
Fan Total Pressure					0.5851
Stack Exit	41.3	1,334.1	0.0682	15.18	0.1009

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6323HUX (18 x 10 feet) - 7.82 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,149.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.9 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.115	Water Loading	6.72 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	34.14
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.997	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	43,356.5 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.947	Dry Air Rate	2,856.9 lb/min/cell
L/G Ratio	1.816	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	8.4 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	397.4	0.0692	14.80	0.0091
Rain Zone	171.0	494.6	0.0692	14.80	0.0249
Fill	171.0	500.8	0.0687	14.99	0.1779
Spray Zone	171.0	507.1	0.0683	14.99	0.0708
Drift Eliminator	171.0	507.1	0.0683	15.18	0.0271
Fan Inlet	57.9	1,497.6	0.0683	15.18	0.0636
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3724
Net Fan VP	57.9	1,497.6	0.0683	15.18	0.1272
Fan Total Pressure					0.4995
Stack Exit	59.7	1,451.7	0.0683	15.18	0.1195

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7R5GO4

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 10,073.8 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 4.90 ft
Total Fill Height = 4.9 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.15 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 40.00 ft
No. of Cells = 4
Inlet Height = 4.00 ft
Cell Length = 20.00 ft
Cell Width = 20.00 ft
Net Total Inlet Area = 608.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.85 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 17.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 103.3 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	6.63 gpm/ft ²
CF-1200MABT - 4.90 ft	1.545	Pressure Ratio	7.67
Fill Total	1.545	L/G * KaV/L	3.0
Rain Zone	0.067	Fan:Box Ratio	39.4 %
Total KaV/L (CTI)	1.726	Fan Coverage	98.5 %
KaV/L Derate (%)	5	Airflow @ Fan	178,715.0 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	67.58 %
KaV/L Adjusted	1.640	Dry Air Rate	11,694.3 lb/min/cell
L/G Ratio	1.843	Total Fill Ht.	4.9 ft
		Fill Area	380.0 ft ² /cell
		Evaporation Rate	0.80 %
		Evaporation Rate	82.7 gpm (tower)
		Inlet DBT	91.3 °F
		Exit WBT	96.3 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	608.0	1,139.3	0.0692	14.81	0.0746
Rain Zone	1,520.0	455.7	0.0692	14.81	0.0221
Fill	1,520.0	463.0	0.0686	15.05	0.3772
Spray Zone	1,520.0	470.3	0.0680	15.05	0.0754
Drift Eliminator	1,520.0	470.3	0.0680	15.28	0.0228
Fan Inlet	610.5	1,170.8	0.0680	15.28	0.0387
Regain					0.0000
Bouyancy					-0.0032
Sum Static dp					0.6076
Net Fan VP	610.5	1,170.8	0.0680	15.28	0.0774
Fan Total Pressure					0.6850
Stack Exit	630.5	1,133.8	0.0680	15.28	0.0726

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6173HUX (14 x 7 feet) - 2.32 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 615.9 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 14.00 ft
No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 7.00 ft
Cell Width = 7.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 5.00 ft
Seal Disk/Hub Diameter = 0.90 ft
Fan Tip Clearance = 0.75 in
Plenum Hole Diameter = 5.13 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 109.4 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	6.62 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	63.81
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	42.1 %
Total KaV/L (CTI)	0.996	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	23,351.6 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	53.34 %
KaV/L Adjusted	0.947	Dry Air Rate	1,538.8 lb/min/cell
L/G Ratio	1.815	Total Fill Ht.	1.96 ft
		Fill Area	46.6 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	4.5 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	285.4	0.0692	14.80	0.0047
Rain Zone	93.1	489.3	0.0692	14.80	0.0245
Fill	93.1	495.5	0.0687	14.99	0.1738
Spray Zone	93.1	501.6	0.0683	14.99	0.0691
Drift Eliminator	93.1	501.6	0.0683	15.18	0.0264
Fan Inlet	40.0	1,168.0	0.0683	15.18	0.0387
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3364
Net Fan VP	40.0	1,168.0	0.0683	15.18	0.0773
Fan Total Pressure					0.4137
Stack Exit	41.3	1,132.0	0.0683	15.18	0.0727

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6303HUX (18 x 10 feet) - 6.52 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 7.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,078.7 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 105.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.106	Water Loading	6.31 gpm/ft ²
CF-1200MABT - 1.96 ft	0.819	Pressure Ratio	33.83
Fill Total	0.819	L/G * KaV/L	1.7
Rain Zone	0.067	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.991	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	39,465.3 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.942	Dry Air Rate	2,601.0 lb/min/cell
L/G Ratio	1.805	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	7.6 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	361.8	0.0692	14.80	0.0075
Rain Zone	171.0	450.3	0.0692	14.80	0.0212
Fill	171.0	455.9	0.0687	14.99	0.1461
Spray Zone	171.0	461.6	0.0683	14.99	0.0577
Drift Eliminator	171.0	461.6	0.0683	15.17	0.0219
Fan Inlet	57.9	1,363.2	0.0683	15.17	0.0527
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3062
Net Fan VP	57.9	1,363.2	0.0683	15.17	0.1054
Fan Total Pressure					0.4116
Stack Exit	59.7	1,321.4	0.0683	15.17	0.0990

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7M1GOL2

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 91.40 °F
Flow Rate = 8,805.8 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 16.20 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 6.86 ft
Total Fill Height = 6.86 ft
Rain Height = 4.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 12.61 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 56.00 ft
No. of Cells = 2
Inlet Height = 4.50 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 718.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 18.00 ft
Seal Disk/Hub Diameter = 3.24 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 18.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.89 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 102.4 %

KaV/L Data

Spray Zone	0.125
HTP-25 - 6.86 ft	1.134
Fill Total	1.134
Rain Zone	0.077
Total KaV/L (CTI)	1.336
KaV/L Derate (%)	5
HWT Correction (%)	2.3
KaV/L Adjusted	1.239
L/G Ratio	1.518

Thermal Data - Misc.

Water Loading	5.91 gpm/ft ²
Pressure Ratio	6.31
L/G * KaV/L	1.9
Fan:Box Ratio	33.1 %
Fan Coverage	86.3 %
Airflow @ Fan	380,690.5 cfm/cell
Effective Fan Eff.	70.27 %
Dry Air Rate	24,572.2 lb/min/cell
Total Fill Ht.	6.86 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	1.38 %
Evaporation Rate	124.7 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	99.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	718.2	1,007.4	0.0695	14.72	0.0586
Rain Zone	1,489.6	485.7	0.0695	14.72	0.0238
Fill	1,489.6	498.4	0.0684	15.11	0.1811
Spray Zone	1,489.6	511.1	0.0674	15.11	0.0789
Drift Eliminator	1,489.6	511.1	0.0674	15.49	0.0272
Fan Inlet	501.9	1,516.9	0.0674	15.49	0.0644
Regain					0.0000
Bouyancy					-0.0055
Sum Static dp					0.4285
Net Fan VP	501.9	1,516.9	0.0674	15.49	0.1288
Fan Total Pressure					0.5573
Stack Exit	518.4	1,468.7	0.0674	15.49	0.1207

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6123HUX (12x6 feet) - 2.09 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 434.4 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 12.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 6.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 136.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft

Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.12 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.4 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.113	Water Loading	6.35 gpm/ft ²
CF-1200MABT - 1.96 ft	0.815	Pressure Ratio	86.44
Fill Total	0.815	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	36.8 %
Total KaV/L (CTI)	0.996	Fan Coverage	100.0 %
KaV/L Derate (%)	5	Airflow @ Fan	17,082.2 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	41.88 %
KaV/L Adjusted	0.946	Dry Air Rate	1,125.6 lb/min/cell
L/G Ratio	1.815	Total Fill Ht.	1.96 ft
		Fill Area	34.2 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	3.3 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	136.8	243.6	0.0692	14.80	0.0034
Rain Zone	68.4	487.2	0.0692	14.80	0.0243
Fill	68.4	493.3	0.0687	14.99	0.1723
Spray Zone	68.4	499.5	0.0683	14.99	0.0685
Drift Eliminator	68.4	499.5	0.0683	15.18	0.0262
Fan Inlet	25.7	1,329.4	0.0683	15.18	0.0501
Regain					0.0000
Bouyancy					-0.0009
Sum Static dp					0.3439
Net Fan VP	25.7	1,329.4	0.0683	15.18	0.1002
Fan Total Pressure					0.4441
Stack Exit	26.5	1,288.6	0.0683	15.18	0.0941

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6266HUX (16 x 9 feet) - 5.84 BHPThermal Conditions

Tower Model = DFC 60HUX
Power = 7.50 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 947.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 16.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 8.00 ft
Cell Width = 9.00 ft
Net Total Inlet Area = 190.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 109.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.117	Water Loading	6.92 gpm/ft ²
CF-1200MABT - 1.96 ft	0.814	Pressure Ratio	42.42
Fill Total	0.814	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	41.5 %
Total KaV/L (CTI)	0.999	Fan Coverage	99.1 %
KaV/L Derate (%)	5	Airflow @ Fan	35,681.9 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.949	Dry Air Rate	2,351.1 lb/min/cell
L/G Ratio	1.820	Total Fill Ht.	1.96 ft
		Fill Area	68.4 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	6.9 gpm (tower)
		Inlet DBT	90.0 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	190.0	366.3	0.0692	14.80	0.0077
Rain Zone	136.8	508.8	0.0692	14.80	0.0262
Fill	136.8	515.2	0.0687	14.99	0.1889
Spray Zone	136.8	521.7	0.0683	14.99	0.0753
Drift Eliminator	136.8	521.7	0.0683	15.18	0.0288
Fan Inlet	57.9	1,232.5	0.0683	15.18	0.0431
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.3691
Net Fan VP	57.9	1,232.5	0.0683	15.18	0.0861
Fan Total Pressure					0.4552
Stack Exit	59.7	1,194.7	0.0683	15.18	0.0809

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L4GO1

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 3,750.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 10.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.00 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 4.90 ft
Total Fill Height = 4.9 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.65 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 319.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.85 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 16.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 107.0 %

KaV/L Data

Spray Zone	0.111
CF-1200MABT - 4.90 ft	1.435
Fill Total	1.435
Rain Zone	0.059
Total KaV/L (CTI)	1.605
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.525
L/G Ratio	2.006

Thermal Data - Misc.

Water Loading	6.85 gpm/ft ²
Pressure Ratio	15.44
L/G * KaV/L	3.1
Fan:Box Ratio	35.6 %
Fan Coverage	87.0 %
Airflow @ Fan	254,489.9 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	16,571.7 lb/min/cell
Total Fill Ht.	4.9 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	0.87 %
Evaporation Rate	34.7 gpm (tower)
Inlet DBT	90.0 °F
Exit WBT	97.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	319.2	765.8	0.0694	14.75	0.0338
Rain Zone	547.2	446.7	0.0694	14.75	0.0197
Fill	547.2	455.9	0.0686	15.05	0.3693
Spray Zone	547.2	465.1	0.0678	15.05	0.0770
Drift Eliminator	547.2	465.1	0.0678	15.36	0.0221
Fan Inlet	198.8	1,280.4	0.0678	15.36	0.0462
Regain					0.0000
Bouyancy					-0.0040
Sum Static dp					0.5642
Net Fan VP	198.8	1,280.4	0.0678	15.36	0.0923
Fan Total Pressure					0.6565
Stack Exit	205.3	1,239.8	0.0678	15.36	0.0865

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L3GOL4 - TKT

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 85.00 °F
Flow Rate = 11,007.0 gpm
Wet Bulb Temperature = 77.00 °F
Relative Humidity = 75.00 %
Range = 20.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 105.00 °F
Approach = 8.00 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 5.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 14.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 4
Inlet Height = 5.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 1,003.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 21.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.0 %

KaV/L Data

Spray Zone	0.137
HTP-25 - 7.84 ft	1.564
Fill Total	1.564
Rain Zone	0.096
Total KaV/L (CTI)	1.798
KaV/L Derate (%)	5
HWT Correction (%)	1.5
KaV/L Adjusted	1.681
L/G Ratio	1.144

Thermal Data - Misc.

Water Loading	5.03 gpm/ft ²
Pressure Ratio	5.83
L/G * KaV/L	1.9
Fan:Box Ratio	35.6 %
Fan Coverage	87.5 %
Airflow @ Fan	288,140.4 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	18,921.9 lb/min/cell
Total Fill Ht.	7.84 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	1.66 %
Evaporation Rate	173.6 gpm (tower)
Inlet DBT	83.5 °F
Exit WBT	95.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,003.2	1,092.1	0.0704	14.47	0.0697
Rain Zone	2,188.8	500.5	0.0704	14.47	0.0251
Fill	2,188.8	513.6	0.0693	14.85	0.2102
Spray Zone	2,188.8	526.6	0.0682	14.85	0.0725
Drift Eliminator	2,188.8	526.6	0.0682	15.23	0.0294
Fan Inlet	795.0	1,449.7	0.0682	15.23	0.0595
Regain					0.0000
Bouyancy					-0.0056
Sum Static dp					0.4608
Net Fan VP	795.0	1,449.7	0.0682	15.23	0.1190
Fan Total Pressure					0.5798
Stack Exit	821.1	1,403.7	0.0682	15.23	0.1116

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1700 TR 39 X 23

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 89.96 °F
Flow Rate = 20,412.0 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 8.44 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.40 °F
Approach = 5.76 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 4.00 ft
Total Fill Height = 4 ft
Rain Height = 5.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 92.00 ft
No. of Cells = 4
Inlet Height = 5.50 ft
Cell Length = 23.00 ft
Cell Width = 39.00 ft
Net Total Inlet Area = 1,369.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 20.00 ft
Seal Disk/Hub Diameter = 3.60 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 20.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 6.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.3 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 4.00 ft	1.120
Fill Total	1.120
Rain Zone	0.086
Total KaV/L (CTI)	1.327
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.260
L/G Ratio	1.722

Thermal Data - Misc.

Water Loading	5.99 gpm/ft ²
Pressure Ratio	5.41
L/G * KaV/L	2.2
Fan:Box Ratio	35.6 %
Fan Coverage	79.0 %
Airflow @ Fan	423,318.2 cfm/cell
Effective Fan Eff.	71.02 %
Dry Air Rate	27,834.6 lb/min/cell
Total Fill Ht.	4 ft
Fill Area	852.2 ft ² /cell
Evaporation Rate	0.76 %
Evaporation Rate	174.8 gpm (tower)
Inlet DBT	91.3 °F
Exit WBT	95.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,369.0	1,204.4	0.0692	14.81	0.0834
Rain Zone	3,408.6	483.7	0.0692	14.81	0.0335
Fill	3,408.6	490.2	0.0687	15.01	0.2267
Spray Zone	3,408.6	496.8	0.0682	15.01	0.0818
Drift Eliminator	3,408.6	496.8	0.0682	15.21	0.0259
Fan Inlet	1,237.0	1,368.9	0.0682	15.21	0.0531
Regain					0.0000
Bouyancy					-0.0027
Sum Static dp					0.5016
Net Fan VP	1,237.0	1,368.9	0.0682	15.21	0.1062
Fan Total Pressure					0.6078
Stack Exit	1,277.7	1,325.3	0.0682	15.21	0.0995

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2450 X 2

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 7,265.0 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
ShockWave 25 = 5.91 ft
Total Fill Height = 5.91 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 12.16 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 2
Inlet Height = 5.00 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 997.5 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.84 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 139.4 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.130	Water Loading 3.12 gpm/ft ²
ShockWave 25 - 5.91 ft 2.038	Pressure Ratio 5.43
Fill Total 2.038	L/G * KaV/L 2.2
Rain Zone 0.090	Fan:Box Ratio 37.4 %
Total KaV/L (CTI) 2.258	Fan Coverage 81.7 %
KaV/L Derate (%) 5	Airflow @ Fan 601,182.3 cfm/cell
HWT Correction (%) 6.6	Effective Fan Eff. 71.90 %
KaV/L Adjusted 1.996	Dry Air Rate 37,435.0 lb/min/cell
L/G Ratio 1.116	Total Fill Ht. 5.91 ft
	Fill Area 1,163.8 ft ² /cell
	Evaporation Rate 2.38 %
	Evaporation Rate 240.7 gpm (tower)
	Inlet DBT 99.0 °F
	Exit WBT 108.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	997.5	1,139.2	0.0680	15.18	0.0733
Rain Zone	2,327.5	488.2	0.0680	15.18	0.0202
Fill	2,327.5	502.4	0.0669	15.62	0.2126
Spray Zone	2,327.5	516.6	0.0659	15.62	0.0643
Drift Eliminator	2,327.5	516.6	0.0659	16.06	0.0272
Fan Inlet	888.1	1,353.9	0.0659	16.06	0.0502
Regain					0.0000
Bouyancy					-0.0065
Sum Static dp					0.4413
Net Fan VP	888.1	1,353.9	0.0659	16.06	0.1003
Fan Total Pressure					0.5417
Stack Exit	917.4	1,310.6	0.0659	16.06	0.0940

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Shapoorji UAE 2050

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 3,632.4 gpm
Wet Bulb Temperature = 91.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 122.00 °F
Approach = 3.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft
Total Fill Height = 6 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 32.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.5 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 6.00 ft	2.120
Fill Total	2.120
Rain Zone	0.069
Total KaV/L (CTI)	2.310
KaV/L Derate (%)	5
HWT Correction (%)	6.6
KaV/L Adjusted	2.042
L/G Ratio	1.153

Thermal Data - Misc.

Water Loading	3.73 gpm/ft ²
Pressure Ratio	7.22
L/G * KaV/L	2.4
Fan:Box Ratio	37.7 %
Fan Coverage	84.7 %
Airflow @ Fan	466,159.2 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	28,967.6 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	2.37 %
Evaporation Rate	96.1 gpm (tower)
Inlet DBT	99.0 °F
Exit WBT	109.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,033.0	0.0680	15.18	0.0602
Rain Zone	972.8	452.0	0.0680	15.18	0.0119
Fill	972.8	465.6	0.0669	15.64	0.2841
Spray Zone	972.8	479.2	0.0658	15.64	0.0557
Drift Eliminator	972.8	479.2	0.0658	16.09	0.0230
Fan Inlet	373.6	1,247.8	0.0658	16.09	0.0425
Regain					0.0000
Bouyancy					-0.0067
Sum Static dp					0.4709
Net Fan VP	373.6	1,247.8	0.0658	16.09	0.0851
Fan Total Pressure					0.5560
Stack Exit	385.9	1,207.9	0.0658	16.09	0.0797

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1750 X 4 Eriphase

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 18,052.0 gpm
Wet Bulb Temperature = 80.96 °F
Relative Humidity = 75.00 %
Range = 25.20 °F
Altitude = 800.0 ft
Hot Water Temperature = 114.80 °F
Approach = 8.64 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 13.25 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 56.00 ft
No. of Cells = 4
Inlet Height = 5.00 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 1,064.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 20.00 ft
Seal Disk/Hub Diameter = 3.60 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 20.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.8 %

KaV/L Data

Spray Zone	0.119
CF-1900SB - 7.00 ft	1.915
Fill Total	1.915
Rain Zone	0.082
Total KaV/L (CTI)	2.116
KaV/L Derate (%)	5
HWT Correction (%)	4.4
KaV/L Adjusted	1.916
L/G Ratio	1.536

Thermal Data - Misc.

Water Loading	6.06 gpm/ft ²
Pressure Ratio	6.13
L/G * KaV/L	2.9
Fan:Box Ratio	40.7 %
Fan Coverage	91.1 %
Airflow @ Fan	371,141.1 cfm/cell
Effective Fan Eff.	71.02 %
Dry Air Rate	23,243.3 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	2.11 %
Evaporation Rate	365.1 gpm (tower)
Inlet DBT	87.8 °F
Exit WBT	106.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,064.0	1,284.9	0.0695	14.70	0.0953
Rain Zone	2,979.2	458.9	0.0695	14.70	0.0250
Fill	2,979.2	478.6	0.0678	15.34	0.3661
Spray Zone	2,979.2	498.3	0.0660	15.34	0.0725
Drift Eliminator	2,979.2	498.3	0.0660	15.97	0.0252
Fan Inlet	1,237.0	1,200.2	0.0660	15.97	0.0395
Regain					0.0000
Bouyancy					-0.0093
Sum Static dp					0.6142
Net Fan VP	1,237.0	1,200.2	0.0660	15.97	0.0790
Fan Total Pressure					0.6933
Stack Exit	1,277.7	1,161.9	0.0660	15.97	0.0741

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 70x35ft Dimensions Fan Deck Height 18ft 4.5ft Air Inlet 5ft Layer Fill

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 14,530.0 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 4.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 2
Inlet Height = 4.50 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 897.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 94.6 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.118	Water Loading 6.24 gpm/ft ²
CF-1900SB - 5.00 ft 1.449	Pressure Ratio 5.62
Fill Total 1.449	L/G * KaV/L 2.4
Rain Zone 0.076	Fan:Box Ratio 37.4 %
Total KaV/L (CTI) 1.643	Fan Coverage 81.1 %
KaV/L Derate (%) 5	Airflow @ Fan 562,307.0 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 71.90 %
KaV/L Adjusted 1.524	Dry Air Rate 36,156.5 lb/min/cell
L/G Ratio 1.573	Total Fill Ht. 5 ft
	Fill Area 1,163.8 ft ² /cell
	Evaporation Rate 1.51 %
	Evaporation Rate 208.2 gpm (tower)
	Inlet DBT 87.4 °F
	Exit WBT 100.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	897.8	1,179.1	0.0698	14.64	0.0806
Rain Zone	2,327.5	454.8	0.0698	14.64	0.0222
Fill	2,327.5	469.0	0.0685	15.09	0.2547
Spray Zone	2,327.5	483.2	0.0672	15.09	0.0711
Drift Eliminator	2,327.5	483.2	0.0672	15.55	0.0239
Fan Inlet	888.1	1,266.4	0.0672	15.55	0.0448
Regain					0.0000
Bouyancy					-0.0078
Sum Static dp					0.4896
Net Fan VP	888.1	1,266.4	0.0672	15.55	0.0895
Fan Total Pressure					0.5791
Stack Exit	917.4	1,225.9	0.0672	15.55	0.0839

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Test MT

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 14,530.0 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 4.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 2
Inlet Height = 4.50 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 897.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 94.6 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.118	Water Loading	6.24 gpm/ft ²
CF-1900SB - 5.00 ft	1.449	Pressure Ratio	5.62
Fill Total	1.449	L/G * KaV/L	2.4
Rain Zone	0.076	Fan:Box Ratio	37.4 %
Total KaV/L (CTI)	1.643	Fan Coverage	81.1 %
KaV/L Derate (%)	5	Airflow @ Fan	562,307.0 cfm/cell
HWT Correction (%)	2.3	Effective Fan Eff.	71.90 %
KaV/L Adjusted	1.524	Dry Air Rate	36,156.5 lb/min/cell
L/G Ratio	1.573	Total Fill Ht.	5 ft
		Fill Area	1,163.8 ft ² /cell
		Evaporation Rate	1.51 %
		Evaporation Rate	208.2 gpm (tower)
		Inlet DBT	87.4 °F
		Exit WBT	100.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	897.8	1,179.1	0.0698	14.64	0.0806
Rain Zone	2,327.5	454.8	0.0698	14.64	0.0222
Fill	2,327.5	469.0	0.0685	15.09	0.2547
Spray Zone	2,327.5	483.2	0.0672	15.09	0.0711
Drift Eliminator	2,327.5	483.2	0.0672	15.55	0.0239
Fan Inlet	888.1	1,266.4	0.0672	15.55	0.0448
Regain					0.0000
Bouyancy					-0.0078
Sum Static dp					0.4896
Net Fan VP	888.1	1,266.4	0.0672	15.55	0.0895
Fan Total Pressure					0.5791
Stack Exit	917.4	1,225.9	0.0672	15.55	0.0839

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 7L3GOL1 3300 CMh, Mahathermal

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 14,530.0 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 5.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 13.25 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 4
Inlet Height = 5.00 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 912.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 19.50 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 96.3 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.112	Water Loading 6.64 gpm/ft ²
CF-1900SB - 7.00 ft 1.723	Pressure Ratio 8.12
Fill Total 1.723	L/G * KaV/L 3.1
Rain Zone 0.079	Fan:Box Ratio 35.6 %
Total KaV/L (CTI) 1.914	Fan Coverage 86.3 %
KaV/L Derate (%) 5	Airflow @ Fan 256,517.1 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 69.17 %
KaV/L Adjusted 1.775	Dry Air Rate 16,367.9 lb/min/cell
L/G Ratio 1.768	Total Fill Ht. 7 ft
	Fill Area 547.2 ft ² /cell
	Evaporation Rate 1.51 %
	Evaporation Rate 211.2 gpm (tower)
	Inlet DBT 87.4 °F
	Exit WBT 102.8 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	912.0	1,050.8	0.0698	14.64	0.0640
Rain Zone	2,188.8	437.9	0.0698	14.64	0.0263
Fill	2,188.8	453.3	0.0684	15.15	0.3362
Spray Zone	2,188.8	468.8	0.0669	15.15	0.0708
Drift Eliminator	2,188.8	468.8	0.0669	15.67	0.0222
Fan Inlet	795.0	1,290.6	0.0669	15.67	0.0463
Regain					0.0000
Bouyancy					-0.0072
Sum Static dp					0.5587
Net Fan VP	795.0	1,290.6	0.0669	15.67	0.0925
Fan Total Pressure					0.6513
Stack Exit	821.1	1,249.6	0.0669	15.67	0.0868

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Zydus 670

Thermal Conditions

Tower Model = 70
Power = 25.00 HP
Cold Water Temperature = 88.16 °F
Flow Rate = 2,130.0 gpm
Wet Bulb Temperature = 82.76 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 97.16 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 239.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 93.0 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.130	Water Loading	6.92 gpm/ft ²
CF-1200MABT - 2.94 ft	1.278	Pressure Ratio	17.70
Fill Total	1.278	L/G * KaV/L	2.2
Rain Zone	0.065	Fan:Box Ratio	35.9 %
Total KaV/L (CTI)	1.473	Fan Coverage	88.8 %
KaV/L Derate (%)	5	Airflow @ Fan	161,166.0 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	65.40 %
KaV/L Adjusted	1.399	Dry Air Rate	10,656.7 lb/min/cell
L/G Ratio	1.541	Total Fill Ht.	2.94 ft
		Fill Area	307.8 ft ² /cell
		Evaporation Rate	0.80 %
		Evaporation Rate	15.9 gpm (tower)
		Inlet DBT	89.7 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	239.4	656.1	0.0694	14.74	0.0248
Rain Zone	307.8	510.3	0.0694	14.74	0.0194
Fill	307.8	517.0	0.0690	14.93	0.2798
Spray Zone	307.8	523.6	0.0685	14.93	0.0864
Drift Eliminator	307.8	523.6	0.0685	15.12	0.0292
Fan Inlet	112.6	1,431.4	0.0685	15.12	0.0583
Regain					0.0000
Bouyancy					-0.0018
Sum Static dp					0.4960
Net Fan VP	112.6	1,431.4	0.0685	15.12	0.1165
Fan Total Pressure					0.6126
Stack Exit	116.3	1,386.2	0.0685	15.12	0.1093

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Mahathermal 3500CMH

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 4,402.3 gpm
Wet Bulb Temperature = 80.60 °F
Relative Humidity = 75.00 %
Range = 16.20 °F
Altitude = 710.0 ft
Hot Water Temperature = 105.80 °F
Approach = 9.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 28.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 28.00 ft
Cell Width = 28.00 ft
Net Total Inlet Area = 372.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 18.00 ft
Seal Disk/Hub Diameter = 3.24 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 18.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 5.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 100.2 %

KaV/L Data

Spray Zone	0.121
CF-1200MABT - 2.94 ft	1.281
Fill Total	1.281
Rain Zone	0.064
Total KaV/L (CTI)	1.467
KaV/L Derate (%)	5
HWT Correction (%)	1.7
KaV/L Adjusted	1.368
L/G Ratio	1.537

Thermal Data - Misc.

Water Loading	5.91 gpm/ft ²
Pressure Ratio	8.00
L/G * KaV/L	2.1
Fan:Box Ratio	33.1 %
Fan Coverage	85.8 %
Airflow @ Fan	366,732.7 cfm/cell
Effective Fan Eff.	70.27 %
Dry Air Rate	23,762.2 lb/min/cell
Total Fill Ht.	2.94 ft
Fill Area	744.8 ft ² /cell
Evaporation Rate	1.37 %
Evaporation Rate	60.4 gpm (tower)
Inlet DBT	87.4 °F
Exit WBT	98.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	372.4	934.0	0.0698	14.64	0.0506
Rain Zone	744.8	467.0	0.0698	14.64	0.0168
Fill	744.8	479.7	0.0687	15.04	0.2384
Spray Zone	744.8	492.4	0.0676	15.04	0.0737
Drift Eliminator	744.8	492.4	0.0676	15.43	0.0251
Fan Inlet	251.0	1,461.3	0.0676	15.43	0.0599
Regain					0.0000
Bouyancy					-0.0059
Sum Static dp					0.4586
Net Fan VP	251.0	1,461.3	0.0676	15.43	0.1198
Fan Total Pressure					0.5785
Stack Exit	259.2	1,414.8	0.0676	15.43	0.1123

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Vikram 1100 X 8 power consumption

Thermal Conditions

Tower Model = 70
Power = 58.38 HP
Cold Water Temperature = 91.40 °F
Flow Rate = 38,745.3 gpm
Wet Bulb Temperature = 85.10 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.40 °F
Approach = 6.30 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 5.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 13.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 192.00 ft
No. of Cells = 8
Inlet Height = 5.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 2,257.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 58.38 HP

KaV/L Data

Spray Zone	0.120
CF-1900SB - 7.00 ft	1.483
Fill Total	1.483
Rain Zone	0.081
Total KaV/L (CTI)	1.684
KaV/L Derate (%)	5
HWT Correction (%)	0.1
KaV/L Adjusted	1.598
L/G Ratio	2.162

Thermal Data - Misc.

Water Loading	8.85 gpm/ft ²
Pressure Ratio	12.43
L/G * KaV/L	3.5
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	286,254.9 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	18,561.0 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	0.79 %
Evaporation Rate	305.8 gpm (tower)
Inlet DBT	92.2 °F
Exit WBT	98.7 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	2,257.2	977.1	0.0690	14.85	0.0547
Rain Zone	4,377.6	503.8	0.0690	14.85	0.0439
Fill	4,377.6	513.5	0.0683	15.14	0.4502
Spray Zone	4,377.6	523.1	0.0676	15.14	0.1030
Drift Eliminator	4,377.6	523.1	0.0676	15.42	0.0287
Fan Inlet	1,590.1	1,440.2	0.0676	15.42	0.0582
Regain					0.0000
Bouyancy					-0.0035
Sum Static dp					0.7354
Net Fan VP	1,590.1	1,440.2	0.0676	15.42	0.1165
Fan Total Pressure					0.8518
Stack Exit	1,642.2	1,394.5	0.0676	15.42	0.1092

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Vikram Solar Compressor 1950 TR

Thermal Conditions

Tower Model = 70
Power = 60.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 3,962.5 gpm
Wet Bulb Temperature = 85.10 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 4.50 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 32.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 425.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 17.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 106.6 %

KaV/L Data

Spray Zone	0.127
CF-1900SB - 5.00 ft	1.854
Fill Total	1.854
Rain Zone	0.069
Total KaV/L (CTI)	2.050
KaV/L Derate (%)	5
HWT Correction (%)	2.3
KaV/L Adjusted	1.901
L/G Ratio	1.141

Thermal Data - Misc.

Water Loading	4.07 gpm/ft ²
Pressure Ratio	6.36
L/G * KaV/L	2.2
Fan:Box Ratio	37.7 %
Fan Coverage	84.7 %
Airflow @ Fan	473,538.8 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	30,631.8 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	1.57 %
Evaporation Rate	66.3 gpm (tower)
Inlet DBT	92.2 °F
Exit WBT	99.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	425.6	1,069.0	0.0690	14.85	0.0655
Rain Zone	972.8	467.7	0.0690	14.85	0.0127
Fill	972.8	477.2	0.0683	15.16	0.2543
Spray Zone	972.8	486.8	0.0675	15.16	0.0600
Drift Eliminator	972.8	486.8	0.0675	15.46	0.0244
Fan Inlet	373.6	1,267.5	0.0675	15.46	0.0450
Regain					0.0000
Bouyancy					-0.0048
Sum Static dp					0.4573
Net Fan VP	373.6	1,267.5	0.0675	15.46	0.0901
Fan Total Pressure					0.5473
Stack Exit	385.9	1,227.1	0.0675	15.46	0.0844

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: AADI HEAT EXCHANGER PVT LTD 720 TR

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 5,503.6 gpm
Wet Bulb Temperature = 83.30 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 6.30 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 54.00 ft
No. of Cells = 3
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 478.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 15.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.4 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.136	Water Loading 5.96 gpm/ft ²
CF-1900SB - 5.00 ft 1.646	Pressure Ratio 8.89
Fill Total 1.646	L/G * KaV/L 2.3
Rain Zone 0.067	Fan:Box Ratio 35.9 %
Total KaV/L (CTI) 1.850	Fan Coverage 88.3 %
KaV/L Derate (%) 5	Airflow @ Fan 168,590.2 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 65.40 %
KaV/L Adjusted 1.715	Dry Air Rate 10,872.9 lb/min/cell
L/G Ratio 1.332	Total Fill Ht. 5 ft
	Fill Area 307.8 ft ² /cell
	Evaporation Rate 1.54 %
	Evaporation Rate 81.0 gpm (tower)
	Inlet DBT 90.3 °F
	Exit WBT 100.1 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	478.8	1,005.9	0.0693	14.77	0.0583
Rain Zone	923.4	521.6	0.0693	14.77	0.0177
Fill	923.4	534.7	0.0684	15.14	0.3261
Spray Zone	923.4	547.7	0.0674	15.14	0.0846
Drift Eliminator	923.4	547.7	0.0674	15.51	0.0317
Fan Inlet	337.8	1,497.3	0.0674	15.51	0.0627
Regain					0.0000
Bouyancy					-0.0038
Sum Static dp					0.5773
Net Fan VP	337.8	1,497.3	0.0674	15.51	0.1254
Fan Total Pressure					0.7027
Stack Exit	348.8	1,450.1	0.0674	15.51	0.1176

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Kreston 4800

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 22,014.4 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 116.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 7.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 16.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 70.00 ft
No. of Cells = 4
Inlet Height = 7.50 ft
Cell Length = 35.00 ft
Cell Width = 35.00 ft
Net Total Inlet Area = 1,995.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 24.00 ft
Seal Disk/Hub Diameter = 4.32 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 24.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 24.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 92.6 %

KaV/L Data

Spray Zone	0.133
HTP-25 - 7.84 ft	1.618
Fill Total	1.618
Rain Zone	0.121
Total KaV/L (CTI)	1.873
KaV/L Derate (%)	5
HWT Correction (%)	5.0
KaV/L Adjusted	1.686
L/G Ratio	1.095

Thermal Data - Misc.

Water Loading	4.73 gpm/ft ²
Pressure Ratio	5.33
L/G * KaV/L	1.8
Fan:Box Ratio	37.4 %
Fan Coverage	82.1 %
Airflow @ Fan	601,867.4 cfm/cell
Effective Fan Eff.	71.90 %
Dry Air Rate	38,428.3 lb/min/cell
Total Fill Ht.	7.84 ft
Fill Area	1,163.8 ft ² /cell
Evaporation Rate	2.29 %
Evaporation Rate	466.0 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	102.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,995.0	1,134.3	0.0695	14.72	0.0743
Rain Zone	4,655.0	486.1	0.0695	14.72	0.0328
Fill	4,655.0	501.7	0.0682	15.19	0.1952
Spray Zone	4,655.0	517.2	0.0669	15.19	0.0657
Drift Eliminator	4,655.0	517.2	0.0669	15.66	0.0277
Fan Inlet	1,776.1	1,355.4	0.0669	15.66	0.0511
Regain					0.0000
Bouyancy					-0.0079
Sum Static dp					0.4389
Net Fan VP	1,776.1	1,355.4	0.0669	15.66	0.1021
Fan Total Pressure					0.5410
Stack Exit	1,834.8	1,312.1	0.0669	15.66	0.0957

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1400 Test

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 4,650.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 75.00 %
Range = 10.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.00 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 319.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 110.2 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.132	Water Loading 8.5 gpm/ft ²
CF-1900SB - 7.00 ft 1.528	Pressure Ratio 15.32
Fill Total 1.528	L/G * KaV/L 3.4
Rain Zone 0.060	Fan:Box Ratio 35.6 %
Total KaV/L (CTI) 1.720	Fan Coverage 86.3 %
KaV/L Derate (%) 5	Airflow @ Fan 314,573.8 cfm/cell
HWT Correction (%) 0.0	Effective Fan Eff. 69.17 %
KaV/L Adjusted 1.634	Dry Air Rate 20,451.4 lb/min/cell
L/G Ratio 2.076	Total Fill Ht. 7 ft
	Fill Area 547.2 ft ² /cell
	Evaporation Rate 0.86 %
	Evaporation Rate 44.2 gpm (tower)
	Inlet DBT 90.0 °F
	Exit WBT 98.1 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	319.2	945.1	0.0694	14.75	0.0515
Rain Zone	547.2	551.3	0.0694	14.75	0.0290
Fill	547.2	563.1	0.0686	15.07	0.5489
Spray Zone	547.2	574.9	0.0677	15.07	0.1238
Drift Eliminator	547.2	574.9	0.0677	15.38	0.0354
Fan Inlet	198.8	1,582.7	0.0677	15.38	0.0704
Regain					0.0000
Bouyancy					-0.0041
Sum Static dp					0.8549
Net Fan VP	198.8	1,582.7	0.0677	15.38	0.1409
Fan Total Pressure					0.9958
Stack Exit	205.3	1,532.5	0.0677	15.38	0.1321

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Vikram 1100cmh X 8, 1300 TR X 8

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 91.40 °F
Flow Rate = 38,745.3 gpm
Wet Bulb Temperature = 85.10 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.40 °F
Approach = 6.30 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Fill Height = 7 ft
Rain Height = 5.50 ft
Total Height Fill + Rain + Spray = 13.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 192.00 ft
No. of Cells = 8
Inlet Height = 5.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 2,257.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 20.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 108.5 %

KaV/L Data

Spray Zone	0.129
CF-1900SB - 7.00 ft	1.481
Fill Total	1.481
Rain Zone	0.082
Total KaV/L (CTI)	1.691
KaV/L Derate (%)	5
HWT Correction (%)	0.1
KaV/L Adjusted	1.605
L/G Ratio	2.166

Thermal Data - Misc.

Water Loading	8.85 gpm/ft ²
Pressure Ratio	12.58
L/G * KaV/L	3.5
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	310,170.5 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	20,110.1 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	0.79 %
Evaporation Rate	331.9 gpm (tower)
Inlet DBT	92.2 °F
Exit WBT	98.8 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	2,257.2	1,058.6	0.0690	14.85	0.0643
Rain Zone	4,377.6	545.9	0.0690	14.85	0.0505
Fill	4,377.6	556.3	0.0683	15.14	0.5366
Spray Zone	4,377.6	566.8	0.0676	15.14	0.1227
Drift Eliminator	4,377.6	566.8	0.0676	15.42	0.0343
Fan Inlet	1,590.1	1,560.5	0.0676	15.42	0.0684
Regain					0.0000
Bouyancy					-0.0036
Sum Static dp					0.8732
Net Fan VP	1,590.1	1,560.5	0.0676	15.42	0.1367
Fan Total Pressure					1.0099
Stack Exit	1,642.2	1,511.0	0.0676	15.42	0.1282

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Vikram Solar

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 91.40 °F
Flow Rate = 4,843.0 gpm
Wet Bulb Temperature = 85.10 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 100.40 °F
Approach = 6.30 °F

Fill Section

Spray Height = 1.25 ft
ShockWave 25 = 5.91 ft
Total Fill Height = 5.91 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.66 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 279.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.84 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 17.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 85.2 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.144	Water Loading 11.8 gpm/ft ²
ShockWave 25 - 5.91 ft 1.277	Pressure Ratio 14.36
Fill Total 1.277	L/G * KaV/L 2.8
Rain Zone 0.061	Fan:Box Ratio 47.5 %
Total KaV/L (CTI) 1.482	Fan Coverage 95.5 %
KaV/L Derate (%) 5	Airflow @ Fan 259,639.7 cfm/cell
HWT Correction (%) 0.1	Effective Fan Eff. 69.17 %
KaV/L Adjusted 1.406	Dry Air Rate 16,882.4 lb/min/cell
L/G Ratio 2.026	Total Fill Ht. 5.91 ft
	Fill Area 410.4 ft ² /cell
	Evaporation Rate 0.79 %
	Evaporation Rate 32.7 gpm (tower)
	Inlet DBT 92.2 °F
	Exit WBT 98.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	279.3	897.8	0.0690	14.85	0.0462
Rain Zone	410.4	611.0	0.0690	14.85	0.0340
Fill	410.4	621.8	0.0684	15.12	0.3894
Spray Zone	410.4	632.7	0.0677	15.12	0.1505
Drift Eliminator	410.4	632.7	0.0677	15.38	0.0435
Fan Inlet	198.8	1,306.3	0.0677	15.38	0.0480
Regain					0.0000
Bouyancy					-0.0032
Sum Static dp					0.7083
Net Fan VP	198.8	1,306.3	0.0677	15.38	0.0960
Fan Total Pressure					0.8043
Stack Exit	205.3	1,264.9	0.0677	15.38	0.0900

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Tanajib 1300 DPC70 Brentwood Shockwave25 CF80MAx

Thermal Conditions

Tower Model = 70
Power = 40.64 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 4,914.0 gpm
Wet Bulb Temperature = 87.80 °F
Relative Humidity = 75.00 %
Range = 10.00 °F
Altitude = 70.0 ft
Hot Water Temperature = 105.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
ShockWave 25 = 5.91 ft
CF-1900SB = 1.00 ft
Total Fill Height = 6.91 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 11.66 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 319.2 ft²

Plenum & Fan Section

Drift Eliminator: CF80MAx
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.84 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 40.64 HP

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.114	Water Loading 8.98 gpm/ft ²
ShockWave 25 - 5.91 ft 1.155	Pressure Ratio 12.47
CF-1900SB - 1.00 ft 0.174	L/G * KaV/L 3.2
Fill Total 1.329	Fan:Box Ratio 35.6 %
Rain Zone 0.057	Fan Coverage 87.0 %
Total KaV/L (CTI) 1.500	Airflow @ Fan 269,938.2 cfm/cell
KaV/L Derate (%) 5	Effective Fan Eff. 69.17 %
HWT Correction (%) 1.5	Dry Air Rate 17,649.5 lb/min/cell
KaV/L Adjusted 1.403	Total Fill Ht. 6.91 ft
L/G Ratio 2.305	Fill Area 547.2 ft ² /cell
	Evaporation Rate 0.88 %
	Evaporation Rate 43.1 gpm (tower)
	Inlet DBT 95.1 °F
	Exit WBT 102.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	319.2	808.8	0.0702	14.63	0.0382
Rain Zone	547.2	471.8	0.0702	14.63	0.0247
Fill	547.2	482.6	0.0694	14.96	0.2775
Spray Zone	547.2	493.3	0.0685	14.96	0.0962
Drift Eliminator	547.2	493.3	0.0685	15.29	0.0393
Fan Inlet	198.8	1,358.1	0.0685	15.29	0.0525
Regain					0.0000
Bouyancy					-0.0043
Sum Static dp					0.5239
Net Fan VP	198.8	1,358.1	0.0685	15.29	0.1049
Fan Total Pressure					0.6288
Stack Exit	205.3	1,315.0	0.0685	15.29	0.0984

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Tanajib 1300 DPC70

Thermal Conditions

Tower Model = 70
Power = 22.67 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 3,900.0 gpm
Wet Bulb Temperature = 87.80 °F
Relative Humidity = 75.00 %
Range = 10.00 °F
Altitude = 70.0 ft
Hot Water Temperature = 105.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Total Fill Height = 7 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 24.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 319.2 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 22.67 HP

KaV/L Data

Spray Zone	0.092
CF-1900SB - 7.00 ft	1.398
Fill Total	1.398
Rain Zone	0.056
Total KaV/L (CTI)	1.546
KaV/L Derate (%)	5
HWT Correction (%)	1.5
KaV/L Adjusted	1.446
L/G Ratio	2.338

Thermal Data - Misc.

Water Loading	7.13 gpm/ft ²
Pressure Ratio	15.21
L/G * KaV/L	3.4
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	211,445.6 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	13,815.1 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	0.88 %
Evaporation Rate	34.2 gpm (tower)
Inlet DBT	95.1 °F
Exit WBT	103.1 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	319.2	633.1	0.0702	14.63	0.0234
Rain Zone	547.2	369.3	0.0702	14.63	0.0170
Fill	547.2	377.9	0.0693	14.97	0.2429
Spray Zone	547.2	386.4	0.0685	14.97	0.0579
Drift Eliminator	547.2	386.4	0.0685	15.31	0.0145
Fan Inlet	198.8	1,063.8	0.0685	15.31	0.0322
Regain					0.0000
Bouyancy					-0.0043
Sum Static dp					0.3835
Net Fan VP	198.8	1,063.8	0.0685	15.31	0.0643
Fan Total Pressure					0.4479
Stack Exit	205.3	1,030.1	0.0685	15.31	0.0603

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: GHV Infra 7K4GOL1

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 3,082.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft
Total Fill Height = 5 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 22.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 22.00 ft
Cell Width = 22.00 ft
Net Total Inlet Area = 292.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 15.00 ft
Seal Disk/Hub Diameter = 2.70 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 15.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 16.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 89.4 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.134	Water Loading 6.7 gpm/ft ²
CF-1900SB - 5.00 ft 1.571	Pressure Ratio 13.09
Fill Total 1.571	L/G * KaV/L 2.3
Rain Zone 0.066	Fan:Box Ratio 37.3 %
Total KaV/L (CTI) 1.770	Fan Coverage 90.5 %
KaV/L Derate (%) 5	Airflow @ Fan 249,881.6 cfm/cell
HWT Correction (%) 2.3	Effective Fan Eff. 68.45 %
KaV/L Adjusted 1.641	Dry Air Rate 16,097.8 lb/min/cell
L/G Ratio 1.416	Total Fill Ht. 5 ft
	Fill Area 459.8 ft ² /cell
	Evaporation Rate 1.53 %
	Evaporation Rate 42.2 gpm (tower)
	Inlet DBT 89.3 °F
	Exit WBT 100.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	292.6	810.0	0.0695	14.72	0.0379
Rain Zone	459.8	515.4	0.0695	14.72	0.0183
Fill	459.8	529.4	0.0684	15.12	0.3224
Spray Zone	459.8	543.5	0.0673	15.12	0.0862
Drift Eliminator	459.8	543.5	0.0673	15.52	0.0311
Fan Inlet	174.9	1,428.4	0.0673	15.52	0.0570
Regain					0.0000
Bouyancy					-0.0054
Sum Static dp					0.5475
Net Fan VP	174.9	1,428.4	0.0673	15.52	0.1141
Fan Total Pressure					0.6616
Stack Exit	180.7	1,383.1	0.0673	15.52	0.1070

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Knexir

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,664.0 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 22.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 22.00 ft
Cell Width = 22.00 ft
Net Total Inlet Area = 292.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 105.3 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.119	Water Loading	5.79 gpm/ft ²
CF-1200MABT - 2.94 ft	1.240	Pressure Ratio	12.21
Fill Total	1.240	L/G * KaV/L	2.2
Rain Zone	0.063	Fan:Box Ratio	32.6 %
Total KaV/L (CTI)	1.423	Fan Coverage	87.3 %
KaV/L Derate (%)	5	Airflow @ Fan	220,763.8 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	67.58 %
KaV/L Adjusted	1.351	Dry Air Rate	14,520.0 lb/min/cell
L/G Ratio	1.602	Total Fill Ht.	2.94 ft
		Fill Area	459.8 ft ² /cell
		Evaporation Rate	0.81 %
		Evaporation Rate	22.6 gpm (tower)
		Inlet DBT	91.3 °F
		Exit WBT	94.9 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	292.6	734.9	0.0692	14.81	0.0310
Rain Zone	459.8	467.6	0.0692	14.81	0.0173
Fill	459.8	473.9	0.0687	15.01	0.2338
Spray Zone	459.8	480.1	0.0682	15.01	0.0729
Drift Eliminator	459.8	480.1	0.0682	15.20	0.0240
Fan Inlet	152.6	1,446.3	0.0682	15.20	0.0593
Regain					0.0000
Bouyancy					-0.0023
Sum Static dp					0.4360
Net Fan VP	152.6	1,446.3	0.0682	15.20	0.1186
Fan Total Pressure					0.5545
Stack Exit	157.6	1,400.6	0.0682	15.20	0.1112

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Test 1

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,664.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.71 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 22.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 22.00 ft
Cell Width = 22.00 ft
Net Total Inlet Area = 292.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 14.00 ft
Seal Disk/Hub Diameter = 2.52 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 14.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.79 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 111.6 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.125	Water Loading	5.79 gpm/ft ²
CF-1200MABT - 1.96 ft	0.891	Pressure Ratio	9.76
Fill Total	0.891	L/G * KaV/L	1.7
Rain Zone	0.064	Fan:Box Ratio	32.6 %
Total KaV/L (CTI)	1.080	Fan Coverage	87.2 %
KaV/L Derate (%)	5	Airflow @ Fan	231,679.2 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	67.58 %
KaV/L Adjusted	1.026	Dry Air Rate	15,310.8 lb/min/cell
L/G Ratio	1.609	Total Fill Ht.	1.96 ft
		Fill Area	459.8 ft ² /cell
		Evaporation Rate	0.80 %
		Evaporation Rate	23.7 gpm (tower)
		Inlet DBT	89.3 °F
		Exit WBT	93.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	292.6	770.4	0.0695	14.72	0.0343
Rain Zone	459.8	490.2	0.0695	14.72	0.0189
Fill	459.8	497.1	0.0690	14.93	0.1729
Spray Zone	459.8	503.9	0.0685	14.93	0.0816
Drift Eliminator	459.8	503.9	0.0685	15.13	0.0268
Fan Inlet	152.6	1,517.8	0.0685	15.13	0.0655
Regain					0.0000
Bouyancy					-0.0025
Sum Static dp					0.3974
Net Fan VP	152.6	1,517.8	0.0685	15.13	0.1310
Fan Total Pressure					0.5284
Stack Exit	157.6	1,469.8	0.0685	15.13	0.1228

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Lotte twin cell

Thermal Conditions

Tower Model = 70
Power = 25.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 4,006.6 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 36.00 ft
No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 359.1 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 101.5 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.127	Water Loading	6.51 gpm/ft ²
CF-1200MABT - 2.94 ft	1.237	Pressure Ratio	10.49
Fill Total	1.237	L/G * KaV/L	2.2
Rain Zone	0.064	Fan:Box Ratio	35.9 %
Total KaV/L (CTI)	1.428	Fan Coverage	88.8 %
KaV/L Derate (%)	5	Airflow @ Fan	159,424.8 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	65.40 %
KaV/L Adjusted	1.356	Dry Air Rate	10,484.4 lb/min/cell
L/G Ratio	1.607	Total Fill Ht.	2.94 ft
		Fill Area	307.8 ft ² /cell
		Evaporation Rate	0.81 %
		Evaporation Rate	32.8 gpm (tower)
		Inlet DBT	91.3 °F
		Exit WBT	95.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	359.1	864.7	0.0692	14.81	0.0430
Rain Zone	615.6	504.4	0.0692	14.81	0.0197
Fill	615.6	511.2	0.0687	15.01	0.2737
Spray Zone	615.6	517.9	0.0682	15.01	0.0859
Drift Eliminator	615.6	517.9	0.0682	15.21	0.0284
Fan Inlet	225.2	1,415.9	0.0682	15.21	0.0568
Regain					0.0000
Bouyancy					-0.0018
Sum Static dp					0.5057
Net Fan VP	225.2	1,415.9	0.0682	15.21	0.1136
Fan Total Pressure					0.6193
Stack Exit	232.5	1,371.3	0.0682	15.21	0.1066

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Lotte

Thermal Conditions

Tower Model = 70
Power = 25.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 2,003.3 gpm
Wet Bulb Temperature = 84.20 °F
Relative Humidity = 75.00 %
Range = 9.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 98.60 °F
Approach = 5.40 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 239.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 102.6 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.128	Water Loading	6.51 gpm/ft ²
CF-1200MABT - 2.94 ft	1.236	Pressure Ratio	17.88
Fill Total	1.236	L/G * KaV/L	2.2
Rain Zone	0.064	Fan:Box Ratio	35.9 %
Total KaV/L (CTI)	1.428	Fan Coverage	88.8 %
KaV/L Derate (%)	5	Airflow @ Fan	161,030.2 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	65.40 %
KaV/L Adjusted	1.357	Dry Air Rate	10,589.8 lb/min/cell
L/G Ratio	1.608	Total Fill Ht.	2.94 ft
		Fill Area	307.8 ft ² /cell
		Evaporation Rate	0.81 %
		Evaporation Rate	16.6 gpm (tower)
		Inlet DBT	91.3 °F
		Exit WBT	95.0 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	239.4	655.1	0.0692	14.81	0.0247
Rain Zone	307.8	509.5	0.0692	14.81	0.0200
Fill	307.8	516.3	0.0687	15.01	0.2795
Spray Zone	307.8	523.2	0.0682	15.01	0.0878
Drift Eliminator	307.8	523.2	0.0682	15.21	0.0290
Fan Inlet	112.6	1,430.2	0.0682	15.21	0.0580
Regain					0.0000
Bouyancy					-0.0018
Sum Static dp					0.4972
Net Fan VP	112.6	1,430.2	0.0682	15.21	0.1159
Fan Total Pressure					0.6131
Stack Exit	116.3	1,385.1	0.0682	15.21	0.1087

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 700 Test 2

Thermal Conditions

Tower Model = 70
Power = 30.00 HP
Cold Water Temperature = 93.20 °F
Flow Rate = 2,642.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 14.40 °F
Altitude = 710.0 ft
Hot Water Temperature = 107.60 °F
Approach = 10.80 °F

Fill Section

Spray Height = 1.25 ft
CF-1200MABT = 2.94 ft
Total Fill Height = 2.94 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 7.69 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 239.4 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.25 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.81 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 13.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 99.9 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.127	Water Loading	8.58 gpm/ft ²
CF-1200MABT - 2.94 ft	1.042	Pressure Ratio	19.52
Fill Total	1.042	L/G * KaV/L	2.3
Rain Zone	0.060	Fan:Box Ratio	35.9 %
Total KaV/L (CTI)	1.229	Fan Coverage	89.2 %
KaV/L Derate (%)	5	Airflow @ Fan	170,632.0 cfm/cell
HWT Correction (%)	2.3	Effective Fan Eff.	65.40 %
KaV/L Adjusted	1.139	Dry Air Rate	10,910.4 lb/min/cell
L/G Ratio	2.000	Total Fill Ht.	2.94 ft
		Fill Area	307.8 ft ² /cell
		Evaporation Rate	1.22 %
		Evaporation Rate	32.3 gpm (tower)
		Inlet DBT	89.3 °F
		Exit WBT	102.3 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	239.4	671.0	0.0695	14.72	0.0260
Rain Zone	307.8	521.9	0.0695	14.72	0.0256
Fill	307.8	538.1	0.0682	15.18	0.3137
Spray Zone	307.8	554.4	0.0670	15.18	0.1097
Drift Eliminator	307.8	554.4	0.0670	15.64	0.0324
Fan Inlet	112.6	1,515.4	0.0670	15.64	0.0639
Regain					0.0000
Bouyancy					-0.0046
Sum Static dp					0.5666
Net Fan VP	112.6	1,515.4	0.0670	15.64	0.1278
Fan Total Pressure					0.6943
Stack Exit	116.3	1,467.7	0.0670	15.64	0.1198

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 2200 Mining Engineering

Thermal Conditions

Tower Model = 70
Power = 40.00 HP
Cold Water Temperature = 86.00 °F
Flow Rate = 5,394.0 gpm
Wet Bulb Temperature = 78.80 °F
Relative Humidity = 75.00 %
Range = 18.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 104.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 6.00 ft
Total Fill Height = 6 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 10.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 478.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.90 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 17.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 106.6 %

KaV/L Data

Spray Zone	0.123
CF-1900SB - 6.00 ft	1.858
Fill Total	1.858
Rain Zone	0.066
Total KaV/L (CTI)	2.047
KaV/L Derate (%)	5
HWT Correction (%)	1.2
KaV/L Adjusted	1.920
L/G Ratio	1.373

Thermal Data - Misc.

Water Loading	4.93 gpm/ft ²
Pressure Ratio	7.44
L/G * KaV/L	2.6
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	266,278.1 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	17,337.8 lb/min/cell
Total Fill Ht.	6 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	1.50 %
Evaporation Rate	86.5 gpm (tower)
Inlet DBT	85.5 °F
Exit WBT	97.7 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	478.8	1,054.1	0.0701	14.56	0.0647
Rain Zone	1,094.4	461.2	0.0701	14.56	0.0149
Fill	1,094.4	473.9	0.0689	14.96	0.3097
Spray Zone	1,094.4	486.6	0.0678	14.96	0.0676
Drift Eliminator	1,094.4	486.6	0.0678	15.36	0.0245
Fan Inlet	397.3	1,340.3	0.0678	15.36	0.0506
Regain					0.0000
Bouyancy					-0.0058
Sum Static dp					0.5263
Net Fan VP	397.3	1,340.3	0.0678	15.36	0.1011
Fan Total Pressure					0.6274
Stack Exit	410.5	1,297.2	0.0678	15.36	0.0947

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: 1250 X 2

Thermal Conditions

Tower Model = 70
Power = 50.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 7,925.2 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 14.40 °F
Altitude = 710.0 ft
Hot Water Temperature = 104.00 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 7.00 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Fill Height = 7 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 11.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 48.00 ft
No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 24.00 ft
Cell Width = 24.00 ft
Net Total Inlet Area = 478.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 16.00 ft
Seal Disk/Hub Diameter = 2.88 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 16.17 ft
Fan Stack Height = 6.00 ft
Regain Height = N/A
Plenum Height = 4.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 18.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 95.3 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 7.00 ft	1.733
Fill Total	1.733
Rain Zone	0.062
Total KaV/L (CTI)	1.916
KaV/L Derate (%)	5
HWT Correction (%)	1.2
KaV/L Adjusted	1.797
L/G Ratio	1.754

Thermal Data - Misc.

Water Loading	7.24 gpm/ft ²
Pressure Ratio	8.63
L/G * KaV/L	3.2
Fan:Box Ratio	35.6 %
Fan Coverage	86.3 %
Airflow @ Fan	276,465.7 cfm/cell
Effective Fan Eff.	69.17 %
Dry Air Rate	17,819.6 lb/min/cell
Total Fill Ht.	7 ft
Fill Area	547.2 ft ² /cell
Evaporation Rate	1.23 %
Evaporation Rate	92.6 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	100.3 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	478.8	1,095.8	0.0695	14.72	0.0693
Rain Zone	1,094.4	479.4	0.0695	14.72	0.0197
Fill	1,094.4	492.3	0.0684	15.12	0.3992
Spray Zone	1,094.4	505.2	0.0673	15.12	0.0837
Drift Eliminator	1,094.4	505.2	0.0673	15.51	0.0265
Fan Inlet	397.5	1,391.0	0.0673	15.51	0.0541
Regain					0.0000
Bouyancy					-0.0054
Sum Static dp					0.6472
Net Fan VP	397.5	1,391.0	0.0673	15.51	0.1082
Fan Total Pressure					0.7554
Stack Exit	410.5	1,346.8	0.0673	15.51	0.1014

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 730 test lo heightThermal Conditions

Tower Model = DFC 70UX
Power = 10.51 HP
Cold Water Temperature = 86.00 °F
Flow Rate = 1,875.0 gpm
Wet Bulb Temperature = 81.00 °F
Relative Humidity = 70.00 %

Range = 10.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 96.00 °F
Approach = 5.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 2.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 2 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 6.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 36.00 ft

No. of Cells = 2
Inlet Height = 3.50 ft
Cell Length = 18.00 ft
Cell Width = 18.00 ft
Net Total Inlet Area = 359.1 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 12.00 ft
Seal Disk/Hub Diameter = 2.16 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 12.17 ft
Fan Stack Height = 4.00 ft
Regain Height = N/A
Plenum Height = 3.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Power: 10.51 HP

KaV/L Data

Spray Zone	0.128
CF-1900SB - 2.00 ft	1.052
Fill Total	1.052
Rain Zone	0.075
Total KaV/L (CTI)	1.254
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	1.191
L/G Ratio	0.846

Thermal Data - Misc.

Water Loading	3.05 gpm/ft ²
Pressure Ratio	5.87
L/G * KaV/L	1.0
Fan:Box Ratio	35.9 %
Fan Coverage	88.3 %
Airflow @ Fan	136,791.9 cfm/cell
Effective Fan Eff.	65.40 %
Dry Air Rate	9,199.7 lb/min/cell
Total Fill Ht.	2 ft
Fill Area	307.8 ft ² /cell
Evaporation Rate	0.99 %
Evaporation Rate	18.7 gpm (tower)
Inlet DBT	89.5 °F
Exit WBT	88.1 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	359.1	753.9	0.0694	14.71	0.0328
Rain Zone	615.6	439.8	0.0694	14.71	0.0088
Fill	615.6	442.1	0.0694	14.79	0.0870
Spray Zone	615.6	444.4	0.0693	14.79	0.0434
Drift Eliminator	615.6	444.4	0.0693	14.87	0.0204
Fan Inlet	225.2	1,214.9	0.0693	14.87	0.0425
Regain					0.0000
Bouyancy					-0.0003
Sum Static dp					0.2346
Net Fan VP	225.2	1,214.9	0.0693	14.87	0.0849
Fan Total Pressure					0.3195
Stack Exit	232.5	1,176.6	0.0693	14.87	0.0797

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** 7A1DOL1 (6 x 6 feet, 3 fill layer) - 2.09 BHPThermal Conditions

Tower Model = DFC 70UX
Power = 3.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 220.8 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %

Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 3.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 3 ft
Rain Height = 3.50 ft

Total Height Fill + Rain + Spray = 7.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 6.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 6.00 ft
Cell Width = 6.00 ft
Net Total Inlet Area = 79.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 4.00 ft
Seal Disk/Hub Diameter = 0.72 ft

Fan Tip Clearance = 0.65 in
Plenum Hole Diameter = 4.11 ft
Fan Stack Height = 1.00 ft
Regain Height = N/A
Plenum Height = 1.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 11.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 113.4 %

KaV/L Data

Spray Zone	0.121
CF-1900SB - 3.00 ft	0.826
Fill Total	0.826
Rain Zone	0.061
Total KaV/L (CTI)	1.009
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	0.958
L/G Ratio	1.833

Thermal Data - Misc.

Water Loading	6.46 gpm/ft ²
Pressure Ratio	123.58
L/G * KaV/L	1.8
Fan:Box Ratio	36.8 %
Fan Coverage	97.8 %
Airflow @ Fan	17,169.4 cfm/cell
Effective Fan Eff.	41.88 %
Dry Air Rate	1,133.1 lb/min/cell
Total Fill Ht.	3 ft
Fill Area	34.2 ft ² /cell
Evaporation Rate	0.69 %
Evaporation Rate	1.7 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	93.6 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	79.8	210.3	0.0691	14.81	0.0025
Rain Zone	34.2	490.7	0.0691	14.81	0.0211
Fill	34.2	496.3	0.0687	14.98	0.1761
Spray Zone	34.2	502.0	0.0684	14.98	0.0873
Drift Eliminator	34.2	502.0	0.0684	15.15	0.0265
Fan Inlet	12.8	1,336.2	0.0684	15.15	0.0507
Regain					0.0000
Bouyancy					-0.0006
Sum Static dp					0.3637
Net Fan VP	12.8	1,336.2	0.0684	15.15	0.1014
Fan Total Pressure					0.4651
Stack Exit	13.3	1,295.2	0.0684	15.15	0.0953

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** Ravi Sir 310 TestThermal Conditions

Tower Model = DFC 70UX
Power = 12.50 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 1,188.8 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 18.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 113.00 °F
Approach = 12.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft

Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 12.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 12.00 ft
Cell Width = 12.00 ft
Net Total Inlet Area = 159.6 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 9.00 ft
Seal Disk/Hub Diameter = 1.62 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 9.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 104.7 %

<u>KaV/L Data</u>	<u>Thermal Data - Misc.</u>
Spray Zone 0.123	Water Loading 8.69 gpm/ft ²
CF-1900SB - 5.00 ft 1.136	Pressure Ratio 47.50
Fill Total 1.136	L/G * KaV/L 2.6
Rain Zone 0.059	Fan:Box Ratio 45.8 %
Total KaV/L (CTI) 1.319	Fan Coverage 96.1 %
KaV/L Derate (%) 5	Airflow @ Fan 76,194.1 cfm/cell
HWT Correction (%) 3.9	Effective Fan Eff. 61.72 %
KaV/L Adjusted 1.201	Dry Air Rate 4,761.6 lb/min/cell
L/G Ratio 2.159	Total Fill Ht. 5 ft
	Fill Area 136.8 ft ² /cell
	Evaporation Rate 1.54 %
	Evaporation Rate 19.2 gpm (tower)
	Inlet DBT 91.0 °F
	Exit WBT 107.3 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	159.6	441.0	0.0692	14.78	0.0112
Rain Zone	136.8	514.5	0.0692	14.78	0.0267
Fill	136.8	535.7	0.0676	15.39	0.3480
Spray Zone	136.8	557.0	0.0660	15.39	0.1128
Drift Eliminator	136.8	557.0	0.0660	16.00	0.0322
Fan Inlet	63.9	1,191.8	0.0660	16.00	0.0389
Regain					0.0000
Bouyancy					-0.0040
Sum Static dp					0.5657
Net Fan VP	63.9	1,191.8	0.0660	16.00	0.0779
Fan Total Pressure					0.6436
Stack Exit	66.0	1,154.5	0.0660	16.00	0.0731

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:** Ravi Sir 285 - 45,35,28, 4500LPMThermal Conditions

Tower Model = DFC 70UX
Power = 12.50 HP
Cold Water Temperature = 95.00 °F
Flow Rate = 1,188.8 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 70.00 %

Range = 18.00 °F
Altitude = 750.0 ft
Hot Water Temperature = 113.00 °F
Approach = 12.60 °F

Fill Section

Spray Height = 1.25 ft
CF-1900SB = 5.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %

Total Fill Height = 5 ft
Rain Height = 3.50 ft
Total Height Fill + Rain + Spray = 9.75 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 11.00 ft

No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 11.00 ft
Cell Width = 11.00 ft
Net Total Inlet Area = 146.3 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 8.00 ft
Seal Disk/Hub Diameter = 1.44 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 8.17 ft
Fan Stack Height = 2.00 ft
Regain Height = N/A
Plenum Height = 2.75 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 14.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 91.9 %

KaV/L Data

Spray Zone	0.128
CF-1900SB - 5.00 ft	1.135
Fill Total	1.135
Rain Zone	0.059
Total KaV/L (CTI)	1.322
KaV/L Derate (%)	5
HWT Correction (%)	3.9
KaV/L Adjusted	1.204
L/G Ratio	2.163

Thermal Data - Misc.

Water Loading	10.34 gpm/ft ²
Pressure Ratio	56.76
L/G * KaV/L	2.6
Fan:Box Ratio	43.3 %
Fan Coverage	96.9 %
Airflow @ Fan	66,830.5 cfm/cell
Effective Fan Eff.	60.49 %
Dry Air Rate	4,175.8 lb/min/cell
Total Fill Ht.	5 ft
Fill Area	115.0 ft ² /cell
Evaporation Rate	1.54 %
Evaporation Rate	16.9 gpm (tower)
Inlet DBT	91.0 °F
Exit WBT	107.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	146.3	421.9	0.0692	14.78	0.0102
Rain Zone	115.0	536.9	0.0692	14.78	0.0288
Fill	115.0	559.2	0.0676	15.39	0.3824
Spray Zone	115.0	581.4	0.0660	15.39	0.1239
Drift Eliminator	115.0	581.4	0.0660	16.00	0.0354
Fan Inlet	50.8	1,316.8	0.0660	16.00	0.0475
Regain					0.0000
Bouyancy					-0.0041
Sum Static dp					0.6241
Net Fan VP	50.8	1,316.8	0.0660	16.00	0.0950
Fan Total Pressure					0.7191
Stack Exit	52.4	1,275.8	0.0660	16.00	0.0892

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DPC 70UX Models (R1)
Delhi, Delhi
India**Project Description:** Pultruded FRP Cooling Tower CTI Series**Comments:**Thermal Conditions

Tower Model = DFC 70UX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 660.0 gpm
Wet Bulb Temperature = 83.00 °F
Relative Humidity = 70.00 %
Range = 7.50 °F
Altitude = 750.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.00 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 3.92 ft
Total Fill Height = 3.92 ft
Rain Height = 3.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 8.67 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 10.00 ft
No. of Cells = 1
Inlet Height = 3.50 ft
Cell Length = 10.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 133.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 100.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 7.00 ft
Seal Disk/Hub Diameter = 1.26 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 7.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 1.83 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 12.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 116.8 %

KaV/L Data

Spray Zone	0.153
HTP-25 - 3.92 ft	0.682
Fill Total	0.682
Rain Zone	0.065
Total KaV/L (CTI)	0.901
KaV/L Derate (%)	5
HWT Correction (%)	0.0
KaV/L Adjusted	0.856
L/G Ratio	1.594

Thermal Data - Misc.

Water Loading	6.95 gpm/ft ²
Pressure Ratio	34.34
L/G * KaV/L	1.4
Fan:Box Ratio	40.3 %
Fan Coverage	87.5 %
Airflow @ Fan	60,481.6 cfm/cell
Effective Fan Eff.	56.79 %
Dry Air Rate	4,009.0 lb/min/cell
Total Fill Ht.	3.92 ft
Fill Area	95.0 ft ² /cell
Evaporation Rate	0.71 %
Evaporation Rate	5.5 gpm (tower)
Inlet DBT	91.7 °F
Exit WBT	92.4 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	133.0	446.4	0.0691	14.81	0.0114
Rain Zone	95.0	625.0	0.0691	14.81	0.0286
Fill	95.0	630.8	0.0688	14.95	0.1728
Spray Zone	95.0	636.6	0.0686	14.95	0.1353
Drift Eliminator	95.0	636.6	0.0686	15.09	0.0446
Fan Inlet	39.1	1,547.2	0.0686	15.09	0.0682
Regain					0.0000
Bouyancy					-0.0005
Sum Static dp					0.4604
Net Fan VP	39.1	1,547.2	0.0686	15.09	0.1363
Fan Total Pressure					0.5968
Stack Exit	40.3	1,499.3	0.0686	15.09	0.1280

Program Input, Page 1**Company:** Delta Cooling Towers Pvt. Ltd.**Project:** DFC 60HUX Models (R5)
Delhi, Delhi
India**Project Description:** Part of the DFC 60UX Series (H)UX - Height Shorter - length and width Taller - height**Comments:** 6323HUXThermal Conditions

Tower Model = DFC 60HUX
Power = 10.00 HP
Cold Water Temperature = 90.00 °F
Flow Rate = 1,497.0 gpm
Wet Bulb Temperature = 82.94 °F
Relative Humidity = 75.00 %

Range = 7.50 °F
Altitude = 800.0 ft
Hot Water Temperature = 97.50 °F
Approach = 7.06 °F

Fill Section

Spray Height = 1.00 ft
CF-1200MABT = 1.96 ft
Total Fill Height = 1.96 ft
Rain Height = 4.00 ft

Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 6.96 ft

Tower Geometry

Cells: Inline
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 18.00 ft

No. of Cells = 2
Inlet Height = 4.00 ft
Cell Length = 9.00 ft
Cell Width = 10.00 ft
Net Total Inlet Area = 212.8 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 6.00 ft
Seal Disk/Hub Diameter = 1.08 ft

Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 6.17 ft
Fan Stack Height = 1.50 ft
Regain Height = N/A
Plenum Height = 2.54 ft
Spray to Drift Height = 0.75 ft
Fan Deck Height = 10.25 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 84.1 %

<u>KaV/L Data</u>		<u>Thermal Data - Misc.</u>	
Spray Zone	0.114	Water Loading	8.75 gpm/ft ²
CF-1200MABT - 1.96 ft	0.811	Pressure Ratio	34.20
Fill Total	0.811	L/G * KaV/L	1.7
Rain Zone	0.068	Fan:Box Ratio	33.2 %
Total KaV/L (CTI)	0.993	Fan Coverage	94.1 %
KaV/L Derate (%)	5	Airflow @ Fan	43,343.1 cfm/cell
HWT Correction (%)	0.0	Effective Fan Eff.	35.87 %
KaV/L Adjusted	0.944	Dry Air Rate	2,856.0 lb/min/cell
L/G Ratio	1.827	Total Fill Ht.	1.96 ft
		Fill Area	85.5 ft ² /cell
		Evaporation Rate	0.67 %
		Evaporation Rate	8.5 gpm (tower)
		Inlet DBT	89.9 °F
		Exit WBT	93.5 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	212.8	397.2	0.0692	14.80	0.0091
Rain Zone	171.0	494.3	0.0692	14.80	0.0250
Fill	171.0	500.6	0.0687	14.99	0.1779
Spray Zone	171.0	506.9	0.0683	14.99	0.0710
Drift Eliminator	171.0	506.9	0.0683	15.18	0.0271
Fan Inlet	57.9	1,497.1	0.0683	15.18	0.0635
Regain					0.0000
Bouyancy					-0.0010
Sum Static dp					0.3726
Net Fan VP	57.9	1,497.1	0.0683	15.18	0.1271
Fan Total Pressure					0.4997
Stack Exit	59.7	1,451.2	0.0683	15.18	0.1194

Program Input, Page 1

Company: Delta Cooling Towers Pvt. Ltd.
Project: New CTI Series DPC - 70 (Gearbox)
Delhi, Delhi
India

Project Description: New CTI Gearbox

Comments: Kreston 4300 X 2

Thermal Conditions

Tower Model = 70
Power = 75.00 HP
Cold Water Temperature = 89.60 °F
Flow Rate = 19,813.0 gpm
Wet Bulb Temperature = 82.40 °F
Relative Humidity = 75.00 %
Range = 27.00 °F
Altitude = 710.0 ft
Hot Water Temperature = 116.60 °F
Approach = 7.20 °F

Fill Section

Spray Height = 1.25 ft
HTP-25 = 7.84 ft
Total Fill Height = 7.84 ft
Rain Height = 7.50 ft
Nozzle Type = DekSpray Nozzle
KaV/L Derate = 5.0 %
dP Derate = 5.0 %
Fill Obstruction = 5.0 %
Total Height Fill + Rain + Spray = 16.59 ft

Tower Geometry

Cells: Back-to-back
End Walls: Both Ends Open
Louver Type: No Louvers
Louver Coefficient: 1.00
Inlet Obstruction = 5.0 %
Tower Length = 64.00 ft
No. of Cells = 4
Inlet Height = 7.50 ft
Cell Length = 32.00 ft
Cell Width = 32.00 ft
Net Total Inlet Area = 1,824.0 ft²

Plenum & Fan Section

Drift Eliminator: DE120
Drift Obstruction = 5.0 %
Fan Inlet Loss Coefficient = 0.50
Transmission Efficiency = 95.0 %
Total Fan Efficiency = 75.0 %
Fan Diameter = 22.00 ft
Seal Disk/Hub Diameter = 3.96 ft
Fan Tip Clearance = 1.00 in
Plenum Hole Diameter = 22.17 ft
Fan Stack Height = 8.00 ft
Regain Height = N/A
Plenum Height = 5.91 ft
Spray to Drift Height = 1.50 ft
Fan Deck Height = 24.00 ft

Program Input, Page 2

Thermal Results

Solve For:
Tower Capability: 92.1 %

KaV/L Data

Spray Zone	0.141
HTP-25 - 7.84 ft	1.614
Fill Total	1.614
Rain Zone	0.122
Total KaV/L (CTI)	1.877
KaV/L Derate (%)	5
HWT Correction (%)	5.0
KaV/L Adjusted	1.689
L/G Ratio	1.099

Thermal Data - Misc.

Water Loading	5.09 gpm/ft ²
Pressure Ratio	6.19
L/G * KaV/L	1.9
Fan:Box Ratio	37.7 %
Fan Coverage	85.6 %
Airflow @ Fan	537,323.4 cfm/cell
Effective Fan Eff.	71.54 %
Dry Air Rate	34,299.7 lb/min/cell
Total Fill Ht.	7.84 ft
Fill Area	972.8 ft ² /cell
Evaporation Rate	2.29 %
Evaporation Rate	417.3 gpm (tower)
Inlet DBT	89.3 °F
Exit WBT	102.7 °F

Pressure Results

Tower Section	Net Area	Velocity	Density	Sp. Vol.	Pres. Drop
Units	Tower - ft ²	ft/min	lb/ft ³	ft ³ /lb	in. wg.
Air Inlet	1,824.0	1,107.4	0.0695	14.72	0.0708
Rain Zone	3,891.2	519.1	0.0695	14.72	0.0368
Fill	3,891.2	535.7	0.0682	15.19	0.2229
Spray Zone	3,891.2	552.3	0.0669	15.19	0.0759
Drift Eliminator	3,891.2	552.3	0.0669	15.67	0.0321
Fan Inlet	1,494.4	1,438.2	0.0669	15.67	0.0575
Regain					0.0000
Bouyancy					-0.0079
Sum Static dp					0.4880
Net Fan VP	1,494.4	1,438.2	0.0669	15.67	0.1150
Fan Total Pressure					0.6029
Stack Exit	1,543.7	1,392.3	0.0669	15.67	0.1077